# FIRST-PRINCIPLES STUDY OF SUPERCONDUCTIVITY IN PRISTINE 1T-$W_2N$ AND JANUS 1T-WTaN/WTaC MONOLAYERS




**A Thesis Submitted for the Partial Fulfillment of the Requirements for the Master of Science (M. Sc.) Degree in Physics**

**Submitted to the**
**the Central Department of Physics**
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**NEPAL**

By

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**[AUGUST, 2026]**

# Declaration

This thesis entitled **"FIRST-PRINCIPLES STUDY OF SUPERCONDUCTIVITY IN PRISTINE 1T-$W_2N$ AND JANUS 1T-WTaN/WTaC MONOLAYERS"** submitted to the Central Department of Physics, Institute of Science and Technology (IoST), Tribhuvan University, Nepal, in partial fulfillment of the requirements for the degree of Master of Science in Physics, has been carried out by me under the supervision of **Prof. Dr. Narayan Prasad Adhikari**, Central Department of Physics, Tribhuvan University.

I hereby declare that this thesis is my original work. It has not been submitted, either in whole or in part, to any other university or institution for the award of any degree or diploma.

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**Ricky Rai**

Date: August 26, 2026

# Recommendation

This is to recommend that **Ricky Rai** has carried out the thesis entitled**"FIRST-PRINCIPLES STUDY OF SUPERCONDUCTIVITY IN PRISTINE 1T-$W_2N$ AND JANUS 1T-WTaN/WTaC MONOLAYERS"**

for the partial fulfillment of the requirements for the degree of Master of Science in Physics under my supervision. To the best of my knowledge, this work has not been submitted to any other university or institution for the award of any academic degree. The candidate has fulfilled all the academic requirements of the Central Department of Physics, Institute of Science and Technology (IoST), Tribhuvan University, Nepal for the submission of this thesis.

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**[ August 26, 2026]**

# Letter of Approval

The thesis entitled **"FIRST-PRINCIPLES STUDY OF SUPERCONDUCTIVITY IN PRISTINE 1T-$W_2N$ AND JANUS 1T-WTaN/WTaC MONOLAYERS"** submitted by **Ricky Rai** to the Central Department of Physics, Institute of Science and Technology (IoST), Tribhuvan University, Kirtipur, Nepal, has been accepted in partial fulfillment of the requirements for the degree of Master of Science in Physics.

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Ricky Rai

[August, 2026]

# Abstract

Two-dimensional (2D) superconductors have attracted considerable interest because of their unique electronic properties and potential applications in nanoscale electronic and quantum devices. In this work, the structural, electronic, dynamical, and superconducting properties of pristine 1T-$W_2N$, Janus 1T-WTaN, and Janus 1T-WTaC monolayers were investigated using first-principles calculations based on Density Functional Theory (DFT) and Density Functional Perturbation Theory (DFPT) implemented in the Quantum ESPRESSO package. The projector augmented-wave (PAW) method and the Perdew–Burke–Ernzerhof (PBE) exchange–correlation functional within the generalized gradient approximation (GGA) were employed throughout the study.

The optimized lattice constants were found to be approximately 2.80 Å, 2.90 Å, and 3.10 Å for pristine 1T-$W_2N$, Janus 1T-WTaN, and Janus 1T-WTaC, respectively. Electronic band structure calculations revealed metallic behavior for all three monolayers. Phonon dispersion analysis confirmed the dynamical stability of pristine 1T-$W_2N$ and Janus 1T-WTaC, while the small imaginary phonon mode in Janus 1T-WTaN was eliminated by applying 1% biaxial tensile strain.

The calculated electron–phonon coupling constants ($\lambda$) are 1.5101, 0.5204, and 0.2622 for pristine 1T-$W_2N$, 1% strained Janus 1T-WTaN, and Janus 1T-WTaC, respectively. Using the Allen–Dynes modified McMillan equation with a Coulomb pseudopotential of $\mu^* = 0.10$, the corresponding superconducting transition temperatures were estimated to be 12.14 K, 2.37 K, and 0.0835 K. The results demonstrate that pristine 1T-$W_2N$ exhibits the strongest electron–phonon coupling and the highest superconducting transition temperature, while Janus engineering reduces the superconducting performance. The successful stabilization of Janus 1T-WTaN through tensile strain further highlights strain engineering as an effective strategy for designing stable Janus superconductors with tunable properties.

**Keywords:** *Two-dimensional superconductors, Janus monolayers, Transition metal nitrides, Density Functional Theory (DFT), Density Functional Perturbation Theory (DFPT), Electron–phonon coupling, Superconducting transition temperature, Quantum ESPRESSO*

# सोधसार

दुई-आयामिक (2D) अतिचालक पदार्थहरूले उत्कृष्ट विद्युतीय गुण तथा नानो-इलेक्ट्रोनिक्स र क्वान्टम प्रविधिमा सम्भावित प्रयोगका कारण पछिल्ला वर्षहरूमा विशेष अनुसन्धानको विषय बनेका छन्। यस शोधकार्यमा पहिलो सिद्धान्त (First-Principles) मा आधारित घनत्व कार्यात्मक सिद्धान्त (Density Functional Theory: DFT) तथा घनत्व कार्यात्मक विक्षोभ सिद्धान्त (Density Functional Perturbation Theory: DFPT) प्रयोग गरी Quantum ESPRESSO सफ्टवेयरमार्फत प्रिन्टिन 1T-$W_2N$, Janus 1T-WTaN तथा Janus 1T-WTaC मोनोलयरहरूको संरचनात्मक, विद्युतीय, जालीय (Dynamical) तथा अतिचालकीय गुणहरूको विस्तृत अध्ययन गरिएको छ। सम्पूर्ण गणनामा Projector Augmented-Wave (PAW) pseudopotential तथा Perdew–Burke–Ernzerhof (PBE) exchange–correlation functional प्रयोग गरिएको छ।

संरचनात्मक अनुकूलनपछि पिन्टिन 1T-$W_2N$, Janus 1T-WTaN तथा Janus 1T-WTaC का सन्तुलित जाली नियताङ्कहरू क्रमशः 2.80 Å, 2.90 Å र 3.10 Å प्राप्त भएका छन्। विद्युतीय ब्यान्ड संरचना तथा घनत्व अवस्थाहरूको विश्लेषणले तीनै मोनोलयर धात्विक प्रकृतिका रहेको देखाएको छ। फोनोन विश्लेषणले पिन्टिन 1T-$W_2N$ र Janus 1T-WTaC गतिशील रूपमा स्थिर रहेको तथा Janus 1T-WTaN मा देखिएको सानो काल्पनिक फोनोन आवृत्ति १% द्विअक्षीय तन्य विकृति लागू गरेपछि हटेको देखाएको छ।

इलेक्ट्रोन–फोनोन युग्मन नियताङ्क ($\lambda$) का मानहरू क्रमशः 1.5101, 0.5204 र 0.2622 प्राप्त भए। $\mu^*$=0.10 का लागि Allen–Dynes संशोधित McMillan समीकरण प्रयोग गर्दा अतिचालकीय संक्रमण तापक्रम ($T_c$) क्रमशः 12.14 K, 2.37 K तथा 0.0835 K प्राप्त भयो। प्राप्त परिणामहरूले पिन्टिन 1T-$W_2N$ सबैभन्दा उच्च अतिचालकीय क्षमता भएको पदार्थ रहेको देखाउँछन्, जबकि Janus संरचना तथा रासायनिक प्रतिस्थापनले इलेक्ट्रोन–फोनोन अन्तरक्रिया कमजोर बनाउँदै $T_c$ घटाएको पाइयो। विशेष रूपमा १% तन्य विकृतिद्वारा Janus 1T-WTaN लाई स्थिर बनाउन सकिने तथ्य यस अध्ययनको महत्वपूर्ण उपलब्धि हो।

# List of Acronyms and Abbreviations

| | |
|---|---|
| **BO** | Born–Oppenheimer |
| **BCS** | Bardeen–Cooper–Schrieffer |
| **DFT** | Density Functional Theory |
| **DFPT** | Density Functional Perturbation Theory |
| **DOS** | Density of States |
| **PDOS** | Projected Density of States |
| **PhDOS** | Phonon Density of States |
| **EPC** | Electron–Phonon Coupling |
| **Eliashberg** | Eliashberg Spectral Function |
| **GGA** | Generalized Gradient Approximation |
| **HF** | Hartree–Fock |
| **KS** | Kohn–Sham |
| **LDA** | Local Density Approximation |
| **PAW** | Projector Augmented-Wave |
| **PBE** | Perdew–Burke–Ernzerhof |
| **QE** | Quantum ESPRESSO |
| **SCF** | Self-Consistent Field |
| **SOC** | Spin–Orbit Coupling |
| **TC** | Superconducting Transition Temperature |
| **XC** | Exchange–Correlation |

# List of Symbols

| | |
|---|---|
| $\lambda$ | Electron–phonon coupling constant |
| $\alpha^2F(\omega)$ | Eliashberg spectral function |
| $\omega_{\log}$ | Logarithmic average phonon frequency |
| $\mu^*$ | Coulomb pseudopotential |
| $T_c$ | Superconducting transition temperature |
| $E_F$ | Fermi energy |
| $N(E_F)$ | Electronic density of states at the Fermi level |

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# Chapter 1

# Introduction

## 1.1 General Consideration

In this chapter, a general introduction to superconductivity is presented, including its history and the different types of superconductors. The chapter also describes the crystal structure, atomic stacking, space group, point group, and other structural properties of the 1T-$W_2C$, Janus 1T-WTaN, and Janus 1T-WTaC monolayers.

## 1.2 Superconductivity

Zero electrical resistance and the total expulsion of magnetic fields (Meissner effect) below a certain temperature ($T_c$) are characteristics of superconductivity, an amazing quantum event. Heike Kamerlingh Onnes made the initial discovery in 1911 when he noticed that mercury's electrical resistance abruptly disappeared at about 4.2 K after chilling it with liquid helium [1]. This discovery initiated extensive research on superconducting materials and their potential technological applications.

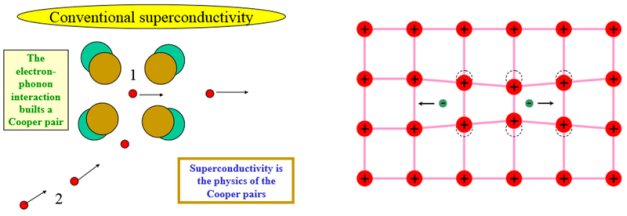



**Figure 1:** Cooper pair

In 1957, the Bardeen–Cooper–Schrieffer (BCS) theory correctly explained the microscopic genesis of conventional superconductivity by attributing superconductivity to the

creation of Cooper pairs via electron–phonon interactions [2]. A more thorough description of superconducting behavior was later provided by Migdal and Eliashberg's extension of the BCS theory to describe materials with enhanced electron–phonon coupling [3, 4]. High-$T_c$ cuprates, iron-based superconductors, and hydrogen-rich superconductors under high pressure have all been developed as a result of ongoing efforts to find materials with higher superconducting transition temperatures.

Low-dimensional systems have received a lot more attention as a result of recent developments in materials science. Quantum confinement and reduced dimensionality drastically alter a material's electrical, vibrational, and superconducting properties as its dimensionality drops from bulk to atomically thin layers. Experimental studies have demonstrated that superconductivity can persist even in single-atomic-layer materials, making two-dimensional (2D) superconductors an important research area for both fundamental physics and future nanoelectronic devices.

Because of their superior metallic conductivity, structural stability, and adjustable electronic properties, transition-metal carbides and nitrides from the MXene family have garnered a lot of attention among other 2D materials. Several superconducting MXenes, such as $W_2C$, $W_2N$, $Mo_2C$, and $Mo_2N$, have been predicted by first-principles calculations, suggesting that electron–phonon coupling is essential to their superconducting behavior [5]. Janus two-dimensional materials, which are created by substituting atoms on one side of a monolayer, have become attractive options for superconducting applications in more recent times. Through chemical substitution and strain engineering, Janus structures' broken out-of-plane symmetry offers more flexibility to customize the electronic structure, phonon spectrum, spin-orbit coupling, and superconducting properties.

Inspired by these advances, the current work uses Density Functional Theory (DFT) and Density Functional Perturbation Theory (DFPT) to study the structural, electrical, vibrational, and superconducting characteristics of 1T-$W_2C$, Janus 1T-WTaN, and Janus 1T-WTaC monolayers. The next sections detail these monolayers' crystal formations and structural properties.

## 1.3 1T Phase of Tungsten Nitride (1T-$W_2N$)

The 1T phase of tungsten nitride (1T-$W_2N$) crystallizes in a trigonal structure with the space group $P\bar{3}m1$ (No. 164), where the nitrogen atom is sandwiched between two tungsten atomic layers, forming a W–N–W stacking sequence. Each tungsten atom is octahedrally coordinated with the nitrogen atom, giving rise to the characteristic 1T crystal structure. The optimized crystal structure of the 1T-$W_2N$ monolayer has attracted attention as a promising two-dimensional superconducting material as in Fig. 2a. The

side view shows the layered W–N–W arrangement, whereas the top view highlights the hexagonal atomic arrangement within the monolayer. In Fig. 2a, the blue spheres denote



**(a)** Side view of 1T-$W_2N$ **(b)** Top view of 1T-$W_2N$

**Figure 2:** 1H phase of Tungsten Nitride

tungsten (W) atoms, whereas the light blue sphere signifies the nitrogen (N) atom. The fundamental unit cell comprises three atoms, which include two tungsten atoms and one nitrogen atom. The monolayer exhibits a trigonal crystalline configuration characterized by the point group $D_{3d}$ and an ABC-type stacking arrangement along the crystallographic $c$-axis. This stratified shape, together with its metallic electrical configuration, renders the 1T-$W_2N$ monolayer an appropriate subject for exploring electron-phonon interactions and superconductivity.

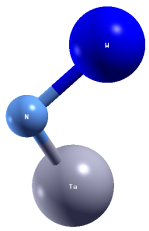
**Table 1:** Optimized fractional atomic coordinates of the 2D 1T-$W_2N$ monolayer.

| Atom | x | y | z |
|---|---|---|---|
| W | 0.3333333964 | 0.6666667928 | 0.4589176289 |
| W | 0.6666666749 | 0.3333331928 | 0.5410823995 |
| N | 0.0000000000 | -0.0000000000 | 0.5000000000 |

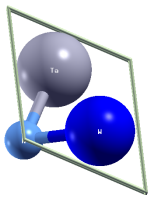
## 1.4 Janus 1T Phase of Tungsten Tantalum Nitride (1T-WTaN)

By substituting one tungsten (W) atomic layer with tantalum (Ta), Janus 1T-WTaN is created from the pristine 1T-$W_2N$ monolayer. This breaks the out-of-plane mirror symmetry while maintaining the trigonal crystal structure. The optimized Janus 1T-WTaN monolayer has the $C_{3v}$ point group and is a member of the $P\bar{3}m1$ (No. 164) space group. Three atoms make up the primitive unit cell: one nitrogen (N), one tantalum (Ta), and one tungsten (W). The distinctive W–N–Ta stacking pattern results from the nitrogen atom being positioned between the tantalum and tungsten layers.

Fig. 3 displays the Janus 1T-WTaN monolayer's optimized crystal structure. While the top image displays the hexagonal atomic arrangement within the two-dimensional lattice, the side view clearly shows the layered W–N–Ta structure.



**(a)** Side view of Janus 1T-WTaN



**(b)** Top view of Janus 1T-WTaN

**Figure 3:** Optimized crystal structure of the Janus 1T-WTaN monolayer.

The side aspect of the optimized Janus 1T-WTaN monolayer is displayed in Figure 3a. Atoms of tungsten (W), tantalum (Ta), and nitrogen (N) are represented by the dark blue, gray, and light blue spheres, respectively. The asymmetric W–N–Ta stacking sequence is formed by the nitrogen atom, which is situated between the tantalum and tungsten layers. In contrast to the pristine 1T-$W_2N$ monolayer, where tungsten atoms occupy both transition-metal layers, the Janus configuration is created when Ta replaces one W layer, breaking the out-of-plane mirror symmetry. The electrical structure, lattice dynamics, and superconducting characteristics are all greatly impacted by this structural imbalance.

The top view of the Janus 1T-WTaN monolayer is shown in Figure 3b. It is evident that the atoms are arranged in a hexagonal pattern inside the two-dimensional lattice. The original unit cell, which has one W atom, one Ta atom, and one N atom, is represented by the green rhombus. When viewed from above, the W and Ta atoms form a periodic hexagonal network despite occupying distinct atomic levels along the out-of-plane direction. Because of the intrinsic out-of-plane electric field created by the asymmetric distribution of transition-metal atoms, Janus 1T-WTaN is a promising material for researching electron-phonon interaction, spin-orbit coupling, and two-dimensional superconductivity.

**Table 2:** Optimized fractional atomic coordinates of the 2D Janus WTaN monolayer.

| Atom | x | y | z |
|---|---|---|---|
| Ta | 0.3333333964 | 0.6666667928 | 0.4598704716 |
| W | 0.6666666749 | 0.3333331928 | 0.5390927375 |
| N | -0.0000000000 | -0.0000000000 | 0.5010368125 |

## 1.5 Janus 1T Phase of Tungsten Tantalum Carbide (1T-WTaC)

The pristine 1T-$W_2C$ monolayer is converted into Janus 1T-WTaC by replacing one tungsten (W) atomic layer with tantalum (Ta). This preserves the trigonal crystal structure while breaking the out-of-plane mirror symmetry. The optimized Janus 1T-WTaC monolayer belongs to the P$\bar{3}$m1 (No. 164) space group and has the $C_{3v}$ point group.. The primordial unit cell consists of one carbon (C), one tantalum (Ta), and one tungsten (W) atom. The carbon atom is positioned between the tantalum and tungsten layers, giving rise to the characteristic W–C–Ta stacking sequence.

Fig. 4 shows the Janus 1T-WTaC monolayer's optimized crystal structure. The top image displays the hexagonal arrangement of atoms within the two-dimensional lattice, but the side view clearly displays the layered W–C–Ta structure.



**(a)** Side view of Janus 1T-WTaC

**(b)** Top view of Janus 1T-WTaC

**Figure 4:** Optimized crystal structure of the Janus 1T-WTaC monolayer.

Figure 4a shows the side view of the optimized Janus 1T-WTaC monolayer. The brown, pink, and dark gray spheres stand in for the atoms of tungsten (W), tantalum (Ta), and carbon (C), respectively. Positioning the carbon atom between the tantalum and tungsten layers results in the asymmetric W–C–Ta stacking sequence. The Janus structure is created and the out-of-plane mirror symmetry is removed when tantalum is used in place of one tungsten layer in pristine 1T-$W_2C$. This structural asymmetry, which modifies the electrical structure, lattice vibrations, and electron-phonon interactions, makes the material a viable choice for superconductivity study.

Figure 4b displays the top view of the optimized Janus 1T-WTaC monolayer. The green rhombus represents the original unit cell, which consists of one W, one Ta, and one C atom. Tantalum and tungsten atoms occupy different atomic layers in the out-of-plane direction while maintaining a periodic hexagonal arrangement in the plane. The asymmetric distribution of transition-metal atoms generates an intrinsic out-of-plane electric

field that can influence the spin-orbit interaction, phonon properties, and superconducting behavior of the Janus monolayer.

**Table 3:** Optimized fractional atomic coordinates of the 2D Janus WTaC monolayer.

| Atom | x | y | z |
|---|---|---|---|
| W | 0.3333536058 | 0.6666834111 | 0.5518411685 |
| Ta | 0.6666624754 | 0.3333323893 | 0.4468496576 |
| C | 0.0000088641 | 0.0000084809 | 0.5013091659 |

## 1.6 Scope of the Present Work

Density Functional Theory (DFT) and Density Functional Perturbation Theory (DFPT) are used in this work to study the structural, electrical, vibrational, and superconducting properties of pristine 1T-$W_2N$, Janus 1T-WTaN, and Janus 1T-WTaC monolayers. Structural optimization, electronic band structure, projected density of states (PDOS), phonon dispersion, projected phonon density of states (PhPDOS), Eliashberg spectral function $\alpha^2F(\omega)$, electron–phonon coupling constant ($\lambda$), and superconducting transition temperature ($T_c$) are all included in the study. The impact of 1% biaxial tensile strain on the Janus 1T-WTaN monolayer's superconducting characteristics and dynamical stability is also examined.

## 1.7 Objectives

### 1.7.1 General Objective

- To investigate the structural, electronic, vibrational, and superconducting properties of pristine 1T-$W_2N$, Janus 1T-WTaN, and Janus 1T-WTaC monolayers using first-principles calculations based on Density Functional Theory (DFT) and Density Functional Perturbation Theory (DFPT).

### 1.7.2 Specific Objectives

- To investigate the electronic properties of pristine $W_2N$ and Janus WTaN and WTaC monolayers using band structure and PDOS calculations.
- To evaluate their dynamical stability and electron–phonon coupling through phonon dispersion, PhPDOS, $\alpha^2F(\omega)$, and $\lambda$ calculations.

- To determine and compare their superconducting transition temperatures ($T_c$), including the effect of $1\%$ biaxial tensile strain on Janus WTaN.

# Chapter 2

# Literature review

## 2.1 General Consideration

In this chapter, history of superconductors, evolution and previous work on two dimensional superconductivity of different families specially Mxene family are reported. Which are essential for the finding the research gap in different kind of system.

## 2.2 Literature Review

Because of its special physical characteristics and potential industrial uses, superconductivity has continued to be one of the most active research fields in condensed matter physics. Understanding the microscopic mechanism of superconductivity and finding materials with greater superconducting transition temperatures have been major priorities since Heike Kamerlingh Onnes discovered it in 1911. The discovery of high-temperature superconductors later expedited the search for new superconducting materials [6–8]. The development of the Bardeen–Cooper–Schrieffer (BCS) theory established electron–phonon interaction as the origin of conventional superconductivity. Because lowering the dimensionality can drastically alter the electrical structure, lattice dynamics, and electron–phonon interaction, two-dimensional (2D) materials have become an intriguing platform for studying superconductivity in recent years. Numerous 2D materials, such as transition metal dichalcogenides (TMDs), MXenes, and BiXenes—many of which display metallic conductivity and superconducting behavior—were extensively studied after graphene was successfully isolated.

Because of its superior electrical conductivity, structural stability, and adjustable surface chemistry, MXenes have garnered a lot of interest among these. Using first-principles

calculations, Bekaert et al. thoroughly examined superconductivity in twenty pristine MXenes and found six superconducting compounds: $Mo_2C$, $W_2C$, $Sc_2C$, $Mo_2N$, $W_2N$, and $Ta_2N$.citebekaert2021a Their findings demonstrated the strong correlation between lattice dynamics and superconductivity [9], with $Mo_2N$ having the highest superconducting transition temperature of about 16 K and $W_2N$ exhibiting competing superconducting and charge-density-wave phases.

Sun *et al.* projected a new class of materials called *BiXenes*, which have a 1H crystal structure different from the typical 1T structure of MXenes, in order to further broaden the family of metallic two-dimensional materials. Several BiXene monolayers are dynamically stable or can be stabilized through strain and finite-temperature effects, according to their first-principles calculations, indicating their potential for electronic and superconducting applications [10].

## 2.3 Research Gap and Motivation

The majority of earlier research has been on symmetric structures and their inherent electrical and superconducting characteristics, despite the fact that superconductivity has been theoretically predicted in a number of pure MXene monolayers, including $W_2N$ and $W_2C$. Specifically, because of its comparatively strong electron–phonon coupling, pristine 1T-$W_2N$ has been discovered as a possible superconducting MXene. However, there hasn't been much focus on how inversion-symmetry breaking through Janus engineering affects tungsten-based nitride MXenes. Similarly, from the standpoint of superconductivity, Janus tungsten-based carbide MXenes are yet mainly investigated.

One transition-metal layer can be substituted to create Janus structures, which are useful for adjusting the electrical structure, lattice dynamics, electron-phonon interaction, and crystal symmetry. A thorough first-principles study of Janus 1T-WTaN and 1T-WTaC monolayers, specifically with regard to their structural stability, electronic characteristics, phonon dispersion, electron–phonon coupling, and superconducting transition temperature, has not yet been published, despite the increasing interest in Janus two-dimensional materials.

Inspired by this research gap, the current work investigates pristine 1T-$W_2N$ together with the Janus 1T-WTaN and 1T-WTaC monolayers using density functional theory (DFT) and density functional perturbation theory (DFPT). To comprehend how transition-metal substitution and inversion-symmetry breaking affect the electron–phonon interaction and superconducting behavior of tungsten-based Janus MXenes, their structural, electronic, vibrational, and superconducting properties are methodically examined.

# Chapter 3

# Materials and Methods

## 3.1 General consideration

The electrical, vibrational, and superconducting characteristics of the 1T-$W_2C$, Janus 1T-WTaN, and Janus 1T-WTaC monolayers were investigated using a variety of theoretical models and approximations. The theoretical foundation and computational techniques employed in this work are described in this chapter.

First, the interacting system of electrons and nuclei is described by the Schrodinger equation and the many-body Hamiltonian. Within the context of the Born–Oppenheimer approximation (BOA) [11], the motion of nuclei may be distinguished from that of electrons since they are significantly heavier than electrons. The Hartree self-consistent field approximation and the Hartree–Fock (HF) approximation, together with their drawbacks, are then presented as two approaches to solve the many-body Schrodinger equation and streamline the description of interacting electrons.

One of the first methods for describing the many-body system of electrons is the Thomas–Fermi (TF) model, which is also briefly reviewed. The ground-state characteristics of interacting electrons in many-body systems are then studied using Density Functional Theory (DFT), a quantum theory. DFT employs the electron density, $\rho(\mathbf{r})$, as the fundamental variable, in contrast to wavefunction-based methods. One might avoid calculating the many-body Schrodinger equation by using the Hohenberg–Kohn theorem, which states that the electron density uniquely determines the ground-state parameters of a many-electron system [12].

A system of non-interacting electrons with the same density but moving in a different effective potential can be used to substitute the many-body interacting system in the Kohn–Sham method [13]. Many approximations for the exchange-correlation functional

are employed because the precise formula for the exchange-correlation energy within DFT is unknown [14]. One such estimate is the Generalized Gradient estimate (GGA), which was proposed by Perdew, Burke, and Ernzerhof (PBE).

The electronic structure calculations, geometry optimization, and calculation of vibrational properties and EPC were performed using the Quantum ESPRESSO package [15, 16]

## 3.2 Many-Body Schrödinger Wave Equation

Atoms, which are made up of positively charged nuclei encircled by electrons, make up all materials. The valence electrons in metallic systems can travel virtually freely throughout the crystal and are only weakly bonded to their parent atoms. It is common practice to treat these movable electrons as an electron gas. The electrons are contained within the crystal lattice and their motion is controlled by the periodic potential formed by the positively charged ion cores. This idea serves as the foundation for the free-electron model, which approximates conduction electrons as a gas of non-interacting particles that follow Fermi-Dirac and the Pauli exclusion principle [19, 20].

Condensed matter systems' electronic, structural, and superconducting characteristics can be comprehended by resolving the Hamiltonian operator's eigenvalue equation, which comprises both the kinetic and potential energy,

$$\hat{H} = \hat{T} + \hat{V}. \tag{3.1}$$

In a real material, electrons and atomic nuclei interact through Coulomb forces, making the system a complex many-body problem. Therefore, the complete quantum mechanical description requires solving the time-independent Schrödinger equation for all interacting particles simultaneously. The Schrödinger equation can be written as

$$\hat{H}\,\Psi(x_1, x_2, \ldots, x_N, R_1, R_2, \ldots, R_M) = E\,\Psi(x_1, x_2, \ldots, x_N, R_1, R_2, \ldots, R_M), \tag{3.2}$$

where $x_i$ denotes the spatial and spin coordinates of the $i$th electron, while $R_a$ represents the position of the $a$th nucleus. The total Hamiltonian of the system is expressed as

$$\hat{H}_{\text{tot}} = \sum_{i=1}^{N} \frac{-\hbar^2}{2m_e}\nabla_i^2 + \sum_{a=1}^{M} \frac{-\hbar^2}{2M_a}\nabla_a^2 + \sum_{i<j}^{N} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} + \sum_{a<b}^{M} \frac{Z_a Z_b e^2}{|\mathbf{R}_a - \mathbf{R}_b|} - \sum_{i=1}^{N}\sum_{a=1}^{M} \frac{Z_a e^2}{|\mathbf{r}_i - \mathbf{R}_a|}. \tag{3.3}$$

In this case, $M_a$ is the mass of the $a$th nucleus, $Z_a$ is its atomic number, and $m_e$ is the mass of the electron. The kinetic energy of electrons and nuclei are represented by the first and second terms, respectively. The fourth term describes the repulsive interaction between nuclei, while the third phrase takes electron-electron Coulomb repulsion into consideration. The attractive Coulomb contact between electrons and nuclei is represented by the last phrase. For systems with a large number of electrons, it is very impossible to achieve a precise analytical solution since the Hamiltonian incorporates interactions between all particles. To make the problem computationally tractable, a number of approximation techniques are used, including Density Functional Theory (DFT) and the Born–Oppenheimer approximation.

## 3.3 Born–Oppenheimer Approximation (BOA)

The Born–Oppenheimer approximation (BOA), also known as the adiabatic approximation [11], is one of the simplest approximations used in quantum mechanics to facilitate the treatment of many-particle systems. The approximation is based on the significant difference between the masses of atomic nuclei and electrons. Because even the smallest nucleus, a proton, is around 1836 times heavier than an electron, nuclei move far more slowly than electrons. Because the electrons can respond to any change in the nuclear positions almost instantaneously, the electronic and nuclear motions can be managed separately. The product of the electronic and nuclear wavefunctions represents the system's overall wavefunction within this approximation.

$$\psi(x, \mathbf{R}) = \psi_e(x, \mathbf{R})\, \psi_n(\mathbf{R}), \tag{3.4}$$

where $\psi_e(x, \mathbf{R})$ represents the electronic wavefunction and $\psi_n(\mathbf{R})$ denotes the nuclear wavefunction.While assuming the nuclear coordinates as fixed parameters, the electronic wavefunction explicitly depends on the electronic coordinates. This approximation is commonly referred to as the zeroth-order Born–Oppenheimer approximation.

Because the kinetic energy associated with the nuclei is much smaller than that of the electrons, the nuclear kinetic energy operator can be neglected during the electronic structure calculation. As a result, the electronic Hamiltonian takes the form

$$\hat{H}_{el} = \sum_{i=1}^{N} \frac{-\hbar^2}{2m_e} \nabla_i^2 + \sum_{i<j}^{N} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} - \sum_{i=1}^{N} \sum_{a=1}^{M} \frac{Z_a e^2}{|\mathbf{r}_i - \mathbf{R}_a|}, \tag{3.5}$$

where $m_e$ is the electron mass, $Z_a$ is the atomic number of the $a$th nucleus, and $\mathbf{R}_a$ denotes its position. In this Hamiltonian, the nuclei provide a fixed external potential in which the electrons move.

The total electronic energy is obtained as the expectation value of the Hamiltonian,

$$E_{\text{tot}} = \langle \hat{H}_{\text{tot}} \rangle = \langle \hat{T} \rangle + \langle \hat{V}_{ee} \rangle + \langle \hat{V}_{\text{ext}} \rangle, \tag{3.6}$$

where $\hat{T}$ is the kinetic energy operator, $\hat{V}_{ee}$ represents the electron–electron interaction, and $\hat{V}_{\text{ext}}$ is the external potential generated by the nuclei.

The external potential acting on the electrons is given by

$$V_{\text{ext}}(\mathbf{r}) = -\sum_{a} \frac{Z_a e^2}{|\mathbf{r} - \mathbf{R}_a|}, \tag{3.7}$$

which explicitly depends on the positions of the nuclei. Although the Born–Oppenheimer approximation greatly simplifies the many-body Schrödinger equation, the electron–electron interaction still couples the electronic coordinates, making an exact analytical solution impossible for realistic systems. Therefore, additional theoretical approaches, such as the Hartree, Hartree–Fock, and Density Functional Theory (DFT) methods, are employed to solve the electronic structure problem efficiently.

## 3.4 Single Particle Approximation

Although the nuclear and electronic motions are separated by the Born-Oppenheimer approximation, electron-electron interactions make it difficult to solve the many-electron Schrödinger equation exactly. The single-particle approximation, which assumes that each electron moves freely in an effective potential created by the average impact of all other electrons, is suggested as a way to simplify this problem [21].

This approximation reduces the many-body problem to a set of single-electron equations, making electronic structure computations computationally feasible. It provides the basis for wavefunction-based methods such as Density Functional Theory (DFT), which is employed in this work , and the Hartree and Hartree–Fock approaches.

## 3.5 Wave Function Approach

We employ Hartree (1928) and Hartree-Fock [24] approximations to solve the many-body Schrdinger wave equation using single particle approximation based on wave function technique. They are described as follows.

## 3.6 Hartree Self-Consistent Field Approximation

One of the first techniques for solving the many-electron Schröder problem was the Hartree self-consistent field (SCF) approximation [23]. This method assumes that the nuclei and the leftover electrons create an average potential in which each electron moves independently. For a system with $N$ electrons, the Schrödinger equation is expressed as

$$\hat{H}\psi_H(r_1, r_2, \ldots, r_N) = E_H\psi_H(r_1, r_2, \ldots, r_N) \tag{3.8}$$

The total wavefunction is the product of single-electron wavefunctions,

$$\psi_H(r_1, r_2, \ldots, r_N) = \phi_1(r_1)\phi_2(r_2)\cdots\phi_N(r_N). \tag{3.9}$$

The total energy is obtained by minimizing

$$E = \langle\psi_H|\hat{H}|\psi_H\rangle = \left\langle \psi_H \left| \left( -\frac{1}{2}\sum_{i=1}^{N}\nabla_i^2 + \frac{1}{2}\sum_{i\neq j}^{N}\frac{1}{|r_i - r_j|} - \sum_{i=1}^{N}\sum_{I=1}^{M}\frac{Z_I}{|r_i - R_I|} \right) \right| \psi_H \right\rangle, \tag{3.10}$$

subject to the normalization condition

$$\int \psi_i^*\psi_i \, d^3r = 1. \tag{3.11}$$

This leads to the Hartree equations,

$$\left( -\frac{1}{2}\nabla_i^2 + V_{\text{ext}} + V_{\text{SC}} \right)\phi_i(r_i) = \lambda_i\phi_i(r_i), \tag{3.12}$$

where

$$V_{\text{ext}} = \sum_I \frac{Z_I}{|r - R_I|} \tag{3.13}$$

and

$$V_{\text{SC}} = V_H = \sum_{j \neq i} \int \frac{|\phi_j(r_j)|^2}{|r_i - r_j|}\, d\tau_j. \tag{3.14}$$

The Hartree equations are solved self-consistently until convergence is achieved. Although this method accounts for the average Coulomb interaction among electrons, it does not satisfy the antisymmetry requirement of fermionic wavefunctions and neglects exchange and correlation effects. These limitations are overcome by more advanced methods such as the Hartree–Fock approximation and Density Functional Theory (DFT).

## 3.7 Hartree–Fock (HF) Approximation

The Hartree–Fock (HF) approximation improves upon the Hartree method by incorporating the Pauli exclusion principle through the use of an antisymmetric wavefunction. The many-electron wavefunction is expressed as a Slater determinant of single-electron spin orbitals in order to accomplish this. [24, 25]:

$$\psi_{\text{HF}} = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_1(r_1, s_1) & \phi_2(r_1, s_1) & \cdots & \phi_N(r_1, s_1) \\ \phi_1(r_2, s_2) & \phi_2(r_2, s_2) & \cdots & \phi_N(r_2, s_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(r_N, s_N) & \phi_2(r_N, s_N) & \cdots & \phi_N(r_N, s_N) \end{vmatrix} \tag{3.15}$$

where $\phi_i$ denotes the single-electron spin orbital, while $r_i$ and $s_i$ represent the spatial and spin coordinates of the $i^{\text{th}}$ electron, respectively.

Minimization of the total energy leads to the Hartree–Fock self-consistent equations,

$$\left( -\frac{1}{2}\nabla^2 + \sum_{I=1}^{M} \frac{Z_I}{|r - R_I|} + \sum_j \int \frac{\phi_j^*(r')\phi_j(r')}{|r - r'|}\, dr' \right) \phi_i(r) - \sum_{j \neq i} \int \frac{\phi_j^*(r')\phi_i(r')}{|r - r'|} \phi_j(r)\, dr' = \varepsilon_i \phi_i(r). \tag{3.16}$$

The third term represents the Coulomb interaction, whereas the 4th term is the exchange interaction, which arises solely from the antisymmetric nature of the fermionic wave-

function. Although the HF method accurately includes exchange effects, it neglects electron correlation beyond exchange. Consequently, its accuracy is limited for many-electron systems, and more advanced approaches such as Density Functional Theory (DFT) are widely used in first-principles calculations.

## 3.8 Thomas–Fermi (TF) Model

One of the first density-based methods that might be considered the forerunner of the current Density Functional Theory (DFT) is the Thomas–Fermi (TF) model [26, 27]. This method treats electrons as a non-interacting gas and expresses their kinetic energy as a function of electron density [26, 27]; the TF equation is as follows.

$$\frac{5}{3} \cdot \frac{3}{10}(3\pi^2)^{2/3} \left|\rho(\mathbf{r})\right|^{2/3} + v_{\text{ext}}(\mathbf{r}) + \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}\, d\mathbf{r}' - \mu = 0, \tag{3.17}$$

where $\rho(\mathbf{r})$ is the electron density, $v_{\text{ext}}(\mathbf{r})$ is the external potential, and $\mu$ is the chemical potential. Solving this equation yields the ground-state electron density. Although the TF model introduced the concept of describing many-electron systems using electron density, it cannot accurately predict atomic shell structures or chemical bonding. These limitations were later overcome by the Hohenberg–Kohn theory, which established the foundation of modern Density Functional Theory [12].

## 3.9 Density Functional Theory (DFT)

One of the most effective quantum mechanical methods for examining the electronic structure of atoms, molecules, and extended systems of matter is Density Functional Theory (DFT). Hohenberg and Kohn [12] and Kohn and Sham [13] established the contemporary theory of DFT by demonstrating that the many-body Schrodinger equation can be rewritten in terms of the density, $n(\mathbf{r})$, rather than the wavefunction.

The idea of using density to describe many-electron systems was pioneered by Thomas [26] and Fermi [27], and later developed by Dirac [28], who proposed a formal expression for the exchange energy; the approach is therefore known as Thomas–Fermi–Dirac theory. Modern DFT was made possible by the Hohenberg–Kohn theorem and Kohn–Sham equations.

The ground-state electron density can be written as

$$n(\mathbf{r}) = N \int \Psi^*(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N)\Psi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N)\, d\mathbf{r}_2\, d\mathbf{r}_3 \cdots d\mathbf{r}_N, \tag{3.18}$$

where $\Psi$ is the many-electron wavefunction and $N$ is the total number of electrons. DFT greatly lowers the computational complexity of solving many-electron systems because the electron density depends only on three spatial variables. Owing to its accuracy and computational efficiency, DFT has become the standard first-principles method for studying the structural, electronic, vibrational, and superconducting properties of materials.

## 3.10 Hohenberg-Kohn Theorem

The two theorems that constitute the basis of the current DFT were proposed and proven by Hohenberg and Kohn in 1964. Density functional theory is developed by Hohenberg and Kohn as an exact theory of many-body systems.

### 3.10.1 First Hohenberg–Kohn (HK) Theorem

With the exception of an additive constant [12], the ground-state electron density, $n(\mathbf{r})$, uniquely determines the external potential, $v(\mathbf{r})$, according to the first Hohenberg–Kohn theorem. Since the system's Hamiltonian is specified by the external potential, the electron density uniquely determines all ground-state parameters, such as the wavefunction and total energy. Therefore, the electron density serves as the fundamental variable in Density Functional Theory (DFT), replacing the many-electron wavefunction. A many-electron system's total energy can be written as a function of the electron density.

$$E[n] = T[n] + V_{ee}[n] + V_{\text{ext}}[n], \tag{3.19}$$

where $T[n]$ is the kinetic energy functional, $V_{ee}[n]$ is the electron–electron interaction energy, and $V_{\text{ext}}[n]$ represents the interaction between the electrons and the external potential.

The external potential energy is given by

$$V_{\text{ext}}[n] = \int v(\mathbf{r})\, n(\mathbf{r})\, d\mathbf{r}. \tag{3.20}$$

Introducing the universal Hohenberg–Kohn functional,

$$F_{\text{HK}}[n] = T[n] + V_{ee}[n], \tag{3.21}$$

the total energy functional becomes

$$E[n] = F_{\text{HK}}[n] + \int v(\mathbf{r})\, n(\mathbf{r})\, d\mathbf{r}. \tag{3.22}$$

The variational principle states that by decreasing the total energy functional with respect to the electron density, the precise ground-state energy can be found by

$$E_0 = \min_n E[n]. \tag{3.23}$$

The ground-state electron density of an interacting many-electron system determines all of its ground-state properties, according to the first Hohenberg–Kohn theorem, which serves as the theoretical basis for DFT.

### 3.10.2 Second Hohenberg–Kohn (HK) Theorem

The variational principle of Density Functional Theory (DFT) is established by the second Hohenberg–Kohn theorem. It asserts that the precise ground-state energy of the system is represented by the minimum value of the total energy functional, which is minimized by the appropriate ground-state electron density [12].

The constraint that the total number of electrons stays constant is applied during the minimization process,

$$N[n] = \int n(\mathbf{r})\, d\mathbf{r} = N. \tag{3.24}$$

Using the method of Lagrange multipliers, the stationary condition for the energy functional is written as

$$\delta \left[ E[n] - \mu \left( \int n(\mathbf{r})\, d\mathbf{r} - N \right) \right] = 0, \tag{3.25}$$

where $\mu$ is the chemical potential. This leads to the Euler–Lagrange equation,

$$\mu = \frac{\delta E[n]}{\delta n(\mathbf{r})} = v_{\text{ext}}(\mathbf{r}) + \frac{\delta F[n]}{\delta n(\mathbf{r})}, \tag{3.26}$$

where the universal Hohenberg–Kohn functional is denoted by $F[n]$. The HK theorems do not give an explicit form of the universal functional, although establishing the theoretical foundation of DFT. The Kohn-Sham method, which established an efficient non-interacting system to make DFT computations computationally feasible, overcame this restriction. [13].

## 3.11 Kohn–Sham Formalism

According to the Hohenberg–Kohn theorems, a many-electron system's ground-state characteristics are exclusively dictated by its electron density. They do not, however, offer a workable way to achieve this density. Kohn and Sham created an auxiliary system of non-interacting electrons that replicates the same ground-state electron density as the interacting system in order to get around this restriction. [13].

The Hamiltonian of the non-interacting system is given by

$$\hat{H}_s = -\frac{1}{2}\sum_{i=1}^{N}\nabla_i^2 + \sum_{i=1}^{N} v_s(\mathbf{r}), \tag{3.27}$$

where $v_s(\mathbf{r})$ is the effective Kohn–Sham potential. The corresponding kinetic energy is

$$T_s = \left\langle \psi_s \left| -\frac{1}{2}\sum_{i=1}^{N}\nabla_i^2 \right| \psi_s \right\rangle. \tag{3.28}$$

The electron density is obtained from the occupied Kohn–Sham orbitals as

$$n(\mathbf{r}) = \sum_{i=1}^{N}\sum_{s} |\phi_i(\mathbf{r}, s)|^2. \tag{3.29}$$

The total energy functional is expressed as

$$E[n] = T_s[n] + J[n] + E_{xc}[n] + \int v(\mathbf{r})n(\mathbf{r})\, d\mathbf{r}, \tag{3.30}$$

where $J[n]$ is the classical Coulomb (Hartree) energy and $E_{xc}[n]$ is the exchange–correlation energy functional, which includes all many-body effects not accounted for by the other terms.

The effective Kohn–Sham potential is given by

$$v_{\text{eff}}(\mathbf{r}) = v(\mathbf{r}) + \int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}\, d\mathbf{r}' + v_{xc}(\mathbf{r}), \tag{3.31}$$

where

$$v_{xc}(\mathbf{r}) = \frac{\delta E_{xc}[n]}{\delta n(\mathbf{r})}. \tag{3.32}$$

The Kohn–Sham equations are therefore

$$\left(-\frac{1}{2}\nabla^2 + v_{\text{eff}}(\mathbf{r})\right)\phi_i(\mathbf{r}) = \varepsilon_i \phi_i(\mathbf{r}). \tag{3.33}$$

These equations are solved self-consistently by starting with an initial guess for the electron density, constructing the effective potential, solving for the Kohn–Sham orbitals, and updating the electron density until convergence is achieved. The Kohn–Sham formalism forms the theoretical basis of modern first-principles calculations and is implemented in computational packages such as Quantum ESPRESSO, which is employed in this work.

## 3.12 Exchange–Correlation Functional

The exchange-correlation (XC) functional plays a major role in the correctness of the computed electronic characteristics in the Kohn-Sham formulation of Density Functional Theory (DFT). The external potential, the exchange–correlation potential, and the Hartree potential make up the effective Kohn–Sham potential [13]. Among these, the XC potential—whose precise form is unknown—accounts for the intricate many-body effects resulting from electron exchange and correlation.

In the Kohn-Sham formulation of Density Functional Theory (DFT), the exchange-correlation (XC) functional is crucial to the accuracy of the calculated electronic characteristics. The external potential, the exchange–correlation potential, and the Hartree potential make up the effective Kohn–Sham potential [13]. Among these, the XC potential—whose precise form is unknown—accounts for the complicated many-body effects occurring from electron exchange and correlation.

Therefore, in order to determine the exchange-correlation energy, adequate approximations are needed. The two most utilized approximations are the Generalized Gradient Approximation (GGA), which takes into consideration both the electron density and its gradient, and the Local Density Approximation (LDA), which simply depends on the

local electron density. All first-principles calculations in this paper use the Perdew–Burke–Ernzerhof (PBE) variation of GGA [14].

### 3.12.1 Local Density Approximation (LDA)

One of the first and most popular approximations for describing the exchange-correlation (XC) energy in Density Functional Theory (DFT) is the Local Density Approximation (LDA), which makes the assumption that the electron density is locally uniform and approximates the XC energy at each point by that of a homogeneous electron gas (HEG) with the same electron density [12, 13].

The LDA exchange–correlation energy is expressed as

$$E_{xc}^{\mathrm{LDA}}[n] = \int n(\mathbf{r})\, \epsilon_{xc}[n(\mathbf{r})]\, d\mathbf{r}, \tag{3.34}$$

where $\epsilon_{xc}[n(\mathbf{r})]$ is the exchange–correlation energy per electron of a homogeneous electron gas. It consists of exchange and correlation contributions,

$$\epsilon_{xc}[n(\mathbf{r})] = \epsilon_x[n(\mathbf{r})] + \epsilon_c[n(\mathbf{r})]. \tag{3.35}$$

LDA provides reliable results for systems with slowly varying electron density and has been successfully applied to many solids. However, its assumption of locally uniform electron density limits its accuracy for systems with rapidly varying charge distributions, where more advanced approximations such as the Generalized Gradient Approximation (GGA) generally provide better results [21, 22].

### 3.12.2 Generalized Gradient Approximation (GGA)

The description of inhomogeneous systems requires an improvement upon the Local Density Approximation, which only considers a uniform electron density. By including the dependence on the density gradient in addition to the density itself, the Generalized Gradient Approximation is able to describe such systems much more accurately than LDA [[29]].

The Generalized Gradient Approximation (GGA) is given by

$$E_{xc}^{\mathrm{GGA}}[n] = \int \epsilon_{xc}\left[n(\mathbf{r}), |\nabla n(\mathbf{r})|\right]\, d\mathbf{r}, \tag{3.36}$$

where $\epsilon_{xc}$ depends on both the local electron density and its gradient. Similar to LDA, the GGA exchange–correlation energy consists of exchange and correlation contributions,

$$E_{xc}^{\mathrm{GGA}}[n] = E_{x}^{\mathrm{GGA}}[n] + E_{c}^{\mathrm{GGA}}[n]. \tag{3.37}$$

The Perdew–Burke–Ernzerhof (PBE) functional is one of the most widely used GGA functionals because it strikes a reasonable compromise between accuracy and computational efficiency [14]. Since the GGA-PBE functional has been shown to produce accurate structural, electrical, vibrational, and superconducting properties of transition-metal-based materials, it is employed for all first-principles calculations in this study.

## 3.13 Density Functional Perturbation Theory (DFPT)

Density Functional Perturbation Theory (DFPT) is a formalism that extends Density Functional Theory (DFT) to enable the computation of many-body systems' response properties to perturbations, such as atomic displacement, an applied electric field, or dynamical lattice deformation [17].

Without using big supercells, DFPT is utilized to compute the dielectric function, Born effective charges, electron-phonon coupling matrix elements, phonon dispersion, and phonon density of states (PhDOS). The energy of the internuclear interaction within the Born-Oppenheimer approximation is given by

$$E_{\mathrm{el}}(R) = \frac{1}{2}\sum_{I\neq J}\frac{Z_I Z_J}{|R_I - R_J|}. \tag{3.38}$$

At equilibrium, the force acting on each nucleus vanishes,

$$\frac{dE(R)}{dR_I} = 0. \tag{3.39}$$

The vibrational frequencies are obtained from the eigenvalues of the mass-weighted Hessian matrix,

$$\det\left(\frac{1}{\sqrt{M_I M_J}}\frac{d^2E(R)}{dR_I dR_J} - \omega_\lambda^2\right) = 0. \tag{3.40}$$

The first-order derivative of the total energy is evaluated using the Hellmann–Feynman theorem,

$$\frac{dE_\lambda}{d\lambda} = \left\langle \psi_\lambda \left| \frac{dH_\lambda}{d\lambda} \right| \psi_\lambda \right\rangle . \tag{3.41}$$

DFPT calculates the dynamical matrix by taking the second derivatives of the total energy with respect to the displacements. The diagonalization of the dynamical matrix gives the eigenvectors (phonon polarization vectors) and eigenvalues (which are proportional to the squared phonon frequencies). These are needed to study the properties related to lattice vibration such as the electron–phonon coupling. The phonon spectra and electron-phonon coupling of the monolayer materials under consideration have been calculated in this work using the DFPT as implemented in the Quantum ESPRESSO suite..

## 3.14 Theoretical Background

Bardeen, Cooper, and Schrieffer (BCS) developed the microscopic theory of conventional superconductivity, explaining that superconductivity results from the creation of Cooper pairs via electron-phonon interactions [2]. The BCS theory states that below a certain temperature ($T_c$), electrons close to the Fermi surface form bonded pairs mediated by lattice vibrations (phonons), leading to a superconducting state.

Although the BCS theory effectively explains weakly linked superconductors, materials with strong electron-phonon interactions cannot be well described by it. In order to get around this restriction, Migdal [3] and Eliashberg [4] created the Migdal–Eliashberg theory, a more rigorous theoretical framework that provides a quantitative description of superconducting properties and appropriately accounts for strong electron–phonon coupling.

Within the context of electron-phonon-mediated superconductivity, the superconducting characteristics of 1T-$W_2C$, Janus 1T-WTaN, and Janus 1T-WTaC monolayers are examined in this work.The superconducting transition temperature ($T_c$) is calculated using first-principles methods based on Density Functional Theory (DFT), Density Functional Perturbation Theory (DFPT), and the Allen–Dynes modified McMillan equation..

## 3.15 BCS Theory

The (BCS) theory is the first successful microscopic theory of conventional superconductivity [7]. According to this theory, electrons near the Fermi surface experience an effective attractive interaction mediated by lattice vibrations (phonons). At tempera-

tures below the superconducting transition temperature ($T_c$), this interaction leads to the formation of bound electron pairs, known as Cooper pairs. The condensation of these Cooper pairs into a coherent quantum state gives rise to superconductivity.

The superconducting energy gap, which represents the energy required to break a Cooper pair, is given by

$$\Delta = \frac{2\hbar\omega_D}{\exp\left(\frac{2}{N(0)V_0}\right) - 1}, \tag{3.42}$$

where $\omega_D$ is the Debye phonon frequency, $N(0)$ is the electronic density of states at the Fermi level per spin, and $V_0$ is the effective attractive interaction between electrons.

For weak electron–phonon coupling,

$$N(0)V_0 \ll 1, \tag{3.43}$$

the above expression reduces to

$$\Delta = 2\hbar\omega_D \exp\left(-\frac{2}{N(0)V_0}\right), \tag{3.44}$$

It demonstrates that the strength of the electron-phonon interaction has an exponential effect on the superconducting energy gap. As a result, a material's superconducting capabilities can be greatly improved by even a slight increase in coupling strength.

### 3.15.1 General Formulation of the Critical Temperature

The superconducting transition temperature ($T_c$), which is determined by the screened Coulomb interaction, the characteristic phonon frequency, and the strength of the electron-phonon coupling, is the temperature below which a material shows a phase transition from the normal conducting to the superconducting state. According to the BCS theory, superconductivity is caused by the formation of Cooper pairs due to electron-phonon interaction [2].

For weak coupling, the superconducting transition temperature can be approximated by

$$T_c \approx \tilde{\Omega} \exp\left(-\frac{1}{\lambda - \mu^*}\right), \tag{3.45}$$

where $\mu^*$ is the Coulomb pseudopotential, $\lambda$ is the electron–phonon coupling constant, and $\tilde{\Omega}$ is the characteristic phonon frequency, which is usually on the order of the Debye

frequency. Equation (3.45) is valid in the weak-coupling limit ($\lambda \ll 1$) and demonstrates that the superconducting transition temperature increases with stronger electron–phonon coupling [18, 30].

For practical first-principles calculations, especially for materials with moderate and strong electron–phonon coupling, the superconducting transition temperature is more accurately estimated using the Allen–Dynes modified McMillan equation, which is employed in the present work.

### 3.15.2 Electron–Phonon Coupling Constant

The electron–phonon coupling (EPC) constant, $\lambda$, is one of the key parameters that determine the properties of conventional superconductors. It characterizes the strength of the electron-phonon interaction that leads to the formation of Cooper pairs. The superconductors are usually separated into weak- (normal) coupling ($\lambda$), and strong-coupling ($\lambda < 1$) types on the basis of the value of $\lambda > 1$ [18, 31].

The electron-phonon coupling constant is defined as

$$\lambda = 2 \int_0^\infty \frac{\alpha^2 F(\omega)}{\omega} \, d\omega, \tag{3.46}$$

where $\alpha^2 F(\omega)$ is the Eliashberg spectral function and $F(\omega)$ is the phonon density of states. The Eliashberg spectral function describes the contribution of phonons to the electron–phonon interaction.

For conventional superconductors, the superconducting transition temperature can be estimated using the Allen–Dynes modified McMillan equation,

$$T_c = \frac{\omega_{\log}}{1.2} \exp\left[ -\frac{1.04(1+\lambda)}{\lambda - \mu^*(1+0.62\lambda)} \right], \tag{3.47}$$

where $\omega_{\log}$ is the logarithmic average phonon frequency, defined as

$$\omega_{\log} = \exp\left[ \frac{2}{\lambda} \int_0^\infty \frac{\alpha^2 F(\omega) \ln \omega}{\omega} \, d\omega \right], \tag{3.48}$$

and $\mu^*$ is the Coulomb pseudopotential, which accounts for the screened electron–electron repulsion. The Allen–Dynes equation provides reliable estimates of the superconducting transition temperature for materials with moderate electron–phonon coupling and is employed in the present work to evaluate the superconducting properties of 1T-$W_2C$, Janus 1T-WTaN, and Janus 1T-WTaC monolayers.

### 3.15.3 Eliashberg Spectral Function

A key quantity in the theory of conventional superconductivity is the Eliashberg spectral function, $\alpha^2F(\omega)$. It explains how various phonon modes contribute to the electron–phonon coupling and characterizes the interaction between electrons and phonons [18, 31]. The electron-phonon coupling constant and the superconducting transition temperature are largely determined by the Eliashberg function. Mathematically, the Eliashberg spectral function is written as

$$\alpha^2F(\omega), \tag{3.49}$$

where $F(\omega)$ is the phonon density of states (PhDOS) and $\alpha^2(\omega)$ is the electron–phonon coupling strength. Thus, the contribution of phonons of frequency $\omega$ to the electron–phonon interaction is measured by the product $\alpha^2F(\omega)$.

Density Functional Perturbation Theory (DFPT) as implemented in Quantum ESPRESSO is used in this work to derive the Eliashberg spectral function. The electron–phonon coupling constant ($\lambda$), the superconducting transition temperature ($T_c$), and the logarithmic average phonon frequency ($\omega_{\log}$) are then calculated using it.

# Chapter 4

# Computer Tools

In this work, a number of computational tools were used for data analysis, crystal structure visualization, first-principles computations, and graphical display of the results. These tools are described in brief below.

## 4.1 Quantum ESPRESSO

Quantum ESPRESSO (QE), an open-source integrated suite of computer tools for electronic-structure calculations and materials modeling, is based on Density Functional Theory (DFT), plane-wave basis sets, and pseudopotentials [15, 16]. It also computes electron-phonon coupling, phonon properties, and superconducting parameters using Density Functional Perturbation Theory (DFPT). The current work used Quantum ESPRESSO to optimize electronic band structures, density of states (DOS), phonon dispersion curves, phonon density of states (PhDOS), Eliashberg spectral function, electron–phonon coupling constant ($\lambda$), logarithmic average phonon frequency ($\omega_{\log}$), and superconducting.

## 4.2 VESTA

A popular three-dimensional visualization tool for showing atomic models, volumetric data, and crystal structures is called VESTA (Visualization for Electronic and Structural Analysis) [32]. It uses structural data files produced by first-principles calculations to produce excellent graphical representations of crystal structures. The optimal structures of 1T-$W_2C$, Janus 1T-WTaN, and Janus 1T-WTaC monolayers were visualized in this study using VESTA, and publication-quality figures displaying the side and top views

of the optimized geometries were prepared.

## 4.3 XCrySDen

An open-source visualization program called XCrySDen was created to show crystalline materials and outcomes from electronic-structure calculations [33]. It offers interactive viewing of crystal structures, Brillouin zones, reciprocal lattices, and electronic band structures and supports several input and output formats generated by Quantum ESPRESSO. The main purposes of XCrySDen in this work were to confirm the optimal crystal structures and analyze the atomic configurations of the monolayers under investigation.

## 4.4 Jupyter Notebook

An open-source interactive computing environment called Jupyter Notebook integrates text, graphical outputs, and executable code into a single document. It is popular for scientific data processing and visualization and supports Python programming. The data from Quantum ESPRESSO computations was analyzed in the current work using Jupyter Notebook, and Python scripts were created to process electron-phonon coupling results, superconducting parameters, and other computational outputs.

## 4.5 XMGrace

A popular tool for visualizing numerical data from computational simulations is XMGrace, a two-dimensional scientific charting program [34]. It offers a number of graph customization options, such as axes, symbols, legends, line styles, and annotations. Electronic band structures, phonon dispersion curves, density of states (DOS), phonon density of states (PhDOS), Eliashberg spectral functions ($\alpha^2F(\omega)$), and electron–phonon coupling functions were all plotted in this study using XMGrace, which made it possible to clearly present and analyze the computational results.

## 4.6 Computational Details

The exchange-correlation interaction was examined using the Perdew–Burke–Ernzerhof (PBE) version of the Generalized Gradient Approximation (GGA), while the inter-

action between the valence electrons and ionic cores was explained using Projector Augmented-Wave (PAW) pseudopotentials from the PSLibrary. Based on the convergence tests, the kinetic-energy and charge-density cutoffs (`ecutwfc` and `ecutrho`) were found to be 90 Ry and 900 Ry, respectively.

While the Janus 1T-WTaN and Janus 1T-WTaC monolayers were structurally optimized using the same $k$-point mesh, the pristine 1T-$W_2N$ monolayer was optimized using the `vc-relax` computation with a $18 \times 18 \times 1$ Monkhorst–Pack $k$-point mesh. The Marzari–Vanderbilt (cold smearing) strategy (`smearing = 'm-v'`) with a smearing width of 0.02 Ry was employed for all SCF computations. The electronic convergence threshold (`conv_thr`) was chosen at $6.0 \times 10^{-5}$ Ry for pristine 1T-$W_2N$ and $6.0 \times 10^{-10}$ Ry for the Janus monolayers, with charge-density mixing parameters (`mixing_beta`) of 0.40 and 0.30, respectively. A vacuum spacing of around 35 Å and the `assume_isolated = '2D'` correction were used to eliminate false interactions between periodically repeated images.

After structural optimization, non-self-consistent field (NSCF) computations were performed using a denser $36 \times 36 \times 1$ Monkhorst–Pack $k$-point mesh. Electronic band structures were calculated along the high-symmetry path $\Gamma$–M–K–$\Gamma$ of the first Brillouin zone using `nbnd = 21`, where `21` denotes the total number of electronic bands used in the computations.

### 4.6.1 Electron–Phonon Coupling Calculations

The EPC calculations for the pristine 1T-$W_2N$ and Janus 1T-WTaN and 1T-WTaC monolayers were performed in the framework of Density Functional Perturbation Theory (DFPT) using the `ph.x` code of the Quantum ESPRESSO suite. The same calculation procedure was applied to all systems under study in order to assess their lattice dynamics and superconducting properties.

Following the geometry optimization, a dense set of self-consistent field (SCF) calculations was performed for each system using a $48 \times 48 \times 1$ Monkhorst–Pack $k$-point grid to sample the Brillouin zone. Such a dense sampling was required to achieve a proper description of the electronic states at the Fermi level. The electronic states near the Fermi energy were found to be critical for the electron–phonon coupling calculations. For this reason, a $k$-point sampling was used to calculate the coupling parameters with sufficient accuracy. The Marzari–Vanderbilt smearing scheme (`smearing = 'm-v'`) and the interpolated electron–phonon coupling approach (`electron_phonon`

= 'interpolated') were applied to perform the calculations. The electron–phonon coupling parameters were evaluated using a Gaussian broadening of 0.005 Ry and 10 smearing values.

The lattice dynamical properties were calculated using the $3 \times 3 \times 1$ phonon ($q$-point) grid, employing DFPT. For this purpose, we used a convergence threshold of $1.0 \times 10^{-14}$ to accurately describe the dynamical matrices and force constants. The calculated dynamical matrices were converted into real-space interatomic force constants using the `q2r.x` code with the crystal acoustic sum rule to impose translational invariance.

The phonon dispersion relations, phonon density of states (PhDOS), and Eliashberg spectral function, $\alpha^2F(\omega)$, were obtained using the `matdyn.x` computation using the real-space force constants as input. Using 1000 interpolation points, the phonon band structures were computed along the high-symmetry $\Gamma$–M–K–$\Gamma$ path. We computed the electron–phonon coupling constant ($\lambda$) and the logarithmic average phonon frequency ($\omega_{\log}$) using the Eliashberg spectral function. Lastly, the Allen–Dynes modified McMillan formula was used to determine the superconducting transition temperature ($T_c$) of the pristine 1T-$W_2N$ and Janus 1T-WTaN and 1T-WTaC monolayers.

# Chapter 5

# Results and Discussion

## 5.1 General Consideration

First-principles computations for pristine 1T-W $_2$ N, Janus 1T-WTaN, and Janus 1T-WTaC monolayers are presented in this chapter. To ensure that our calculations were accurate, we examined the convergence of the Monkhorst–Pack $k$-point mesh and plane-wave kinetic-energy cutoff before doing physical property calculations. In our computations, we employed the ideal convergence parameters.

We present and discuss our structural, electrical, vibrational, and superconducting property calculations of the monolayers under study. We specifically performed a structural optimization and looked at the electronic band structure, density of states, phonon dispersion relation, electron–phonon coupling, Eliashberg spectral function, and superconducting transition temperature in order to ascertain the effect of Janus asymmetry on the superconducting properties of the monolayers under investigation.

## 5.2 Optimization of parameters

### 5.2.1 Kinetic Energy Cutoff Convergence

One of the most crucial factors in plane-wave Density Functional Theory (DFT) computations is the plane-wave kinetic energy cutoff (`ecutwfc`), which establishes the maximum kinetic energy of the plane waves utilized to expand the electronic wave functions. A lower ground-state energy and a more accurate description of the electronic states result from expanding the basis set by raising the kinetic energy cutoff. Nevertheless, a higher cutoff also raises the computational expense. In order to strike a compromise

between computing efficiency and numerical precision, a suitable cutoff energy must be chosen. When additional increases result in only little variations in the total energy, the cutoff energy is said to have converged.

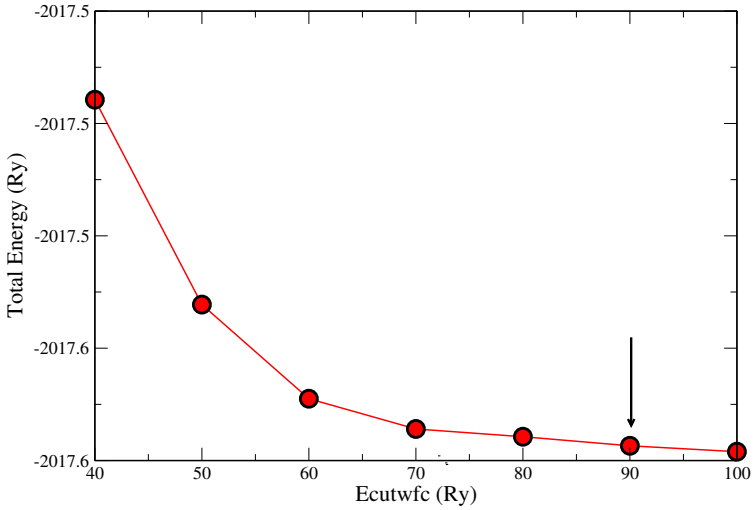



**Figure 5:** Variation of the total energy with respect to the plane-wave kinetic energy cutoff for the pristine 1T-$W_2N$ monolayer.

The convergence of the total energy for the pristine 1T-$W_2N$ monolayer with regard to the plane-wave kinetic energy cutoff is shown in Figure 5. At lower cutoff energies, the total energy is found to decline quickly, and as the cutoff increases, it progressively approaches a constant value. After around 70 Ry, the total energy almost converges. However, a kinetic energy cutoff of 90 Ry was used in order to get greater numerical precision, especially for the ensuing phonon and electron–phonon coupling computations. For pristine 1T-$W_2N$, Janus 1T-WTaN, and Janus 1T-WTaC monolayers, a kinetic energy cutoff of 90 Ry and a charge-density cutoff (`ecutrho`) of 900 Ry were therefore consistently used throughout this work.

### 5.2.2 K-Point Optimization

The sampling of the Brillouin zone has a significant impact on the accuracy of Density Functional Theory computations. A finite set of **k**-points in the first Brillouin zone are used for the Brillouin-zone integration in periodic systems. The Monkhorst–Pack scheme is used in the construction of this set. An improved description of the electronic states and, thus, a more precise total energy result from a greater sampling of **k**-points. Therefore, before moving on to the production computations, a convergence test concerning the **k**-point sampling was carried out.

The total energy as a function of the **k**-point grid is displayed in Fig. 6. The total energy decreases rapidly with increasing **k**-point sampling and reaches a plateau for the $8\times8\times1$ grid. The plane-wave kinetic-energy cutoff for the convergence testing was set to 90 Ry. The total energy was calculated using various Monkhorst–Pack grids.

Although the total energy was found to be converged with respect to the $8\times8\times1$ $k$-point mesh, a denser $18\times18\times1$ Monkhorst–Pack grid was used for the structural optimization and (SCF) calculations in order to obtain the numerical accuracy required for accurate geometry relaxations and electronic structure. We have employed the same $18\times18\times1$ $k$-point mesh for 1T-$W_2N$, Janus 1T-WTaN, and Janus 1T-WTaC monolayers during the calculations of their structural, electronic, vibrational, and superconducting properties in this work to ensure consistent comparison of the results.

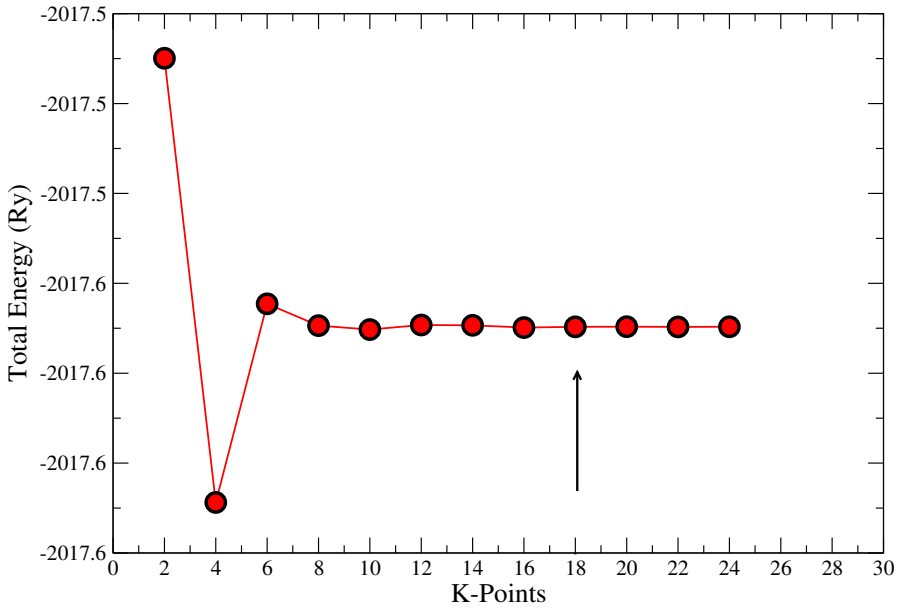
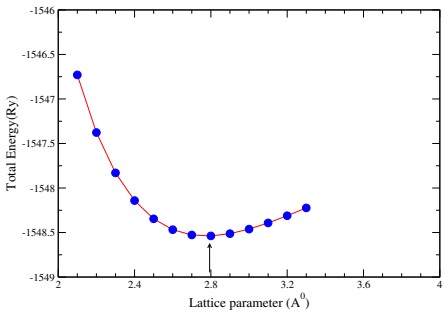



**Figure 6:** Variation of the total energy with different Monkhorst–Pack $k$-point meshes for pristine 1T-$W_2N$.

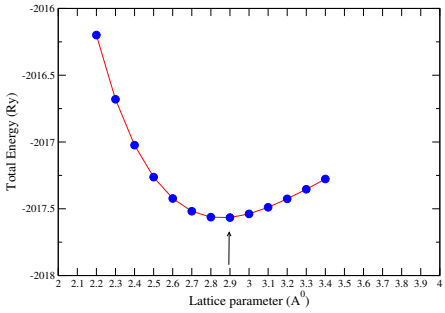
### 5.2.3 Lattice Parameter Optimization

The equilibrium lattice parameters of pristine 1T-$W_2N$ and the Janus 1T-WTaN and 1T-WTaC monolayers were optimized following the determination of the converged plane-wave kinetic energy cutoff and $k$-point mesh. Through a series of self-consistent field (SCF) calculations, the in-plane lattice parameter ($a$=$b$) was systematically varied while all other computational parameters were fixed at their converged values. Only the in-plane lattice constants were optimized because all three systems are two-dimensional hexagonal monolayers; the out-of-plane lattice parameter was left fixed with a big enough vacuum layer to prevent artificial interactions between periodically repeated pictures.

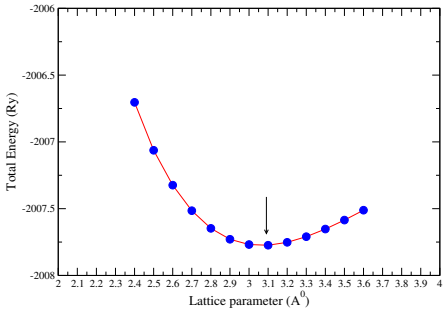
For pristine 1T-$W_2N$, Janus 1T-WTaN, and Janus 1T-WTaC, respectively, Figures 7a–7c show how the total energy changes with the lattice parameter. In each instance, the total energy falls as the lattice parameter gets closer to its equilibrium value, reaches a minimum, and then rises with more expansion. Stronger orbital overlap and more short-range repulsive interactions result from the atoms being pushed closer together at reduced lattice parameters. On the other hand, raising the lattice parameter causes the atoms' chemical bonds to weaken, which raises the overall energy. As a result, the equilibrium lattice parameter at which the crystal reaches its most stable configuration




**(a)** Variation of total energy as a function of lattice parameter for pristine 1T-$W_2N$ monolayer.




**(b)** Variation of total energy as a function of lattice parameter for Janus 1T-WTaN monolayer.




**(c)** Variation of total energy as a function of lattice parameter for Janus 1T-WTaC monolayer.

**Figure 7:** Lattice parameter optimization of pristine 1T-$W_2N$, Janus 1T-WTaN, and Janus 1T-WTaC monolayers.

is represented by the minimum of each energy curve.

The total-energy minimization yielded optimum lattice parameters of roughly 2.80 Å for pristine 1T-$W_2N$, 2.90 Å for Janus 1T-WTaN, and 3.10 Å for Janus 1T-WTaC. The variations in atomic size and bonding properties are primarily responsible for the steady rise in the equilibrium lattice constant following the substitution of one tungsten atom with tantalum and then nitrogen with carbon. All following structural, electronic, phonon, electron-phonon coupling, and superconductivity calculations provided in this paper were performed using these optimized structures.

## 5.3 Structural Properties

The optimized crystal structures of pristine 1T-$W_2N$, Janus 1T-WTaN, and Janus 1T-WTaC monolayers were obtained after variable-cell structural relaxation. The optimized lattice parameters and total energies have already been presented in Table **??**. All subsequent electronic, dynamical, electron–phonon coupling, and superconducting calculations were performed using these optimized equilibrium structures.

The optimized bond lengths calculated from the relaxed structures are summarized in Table 4. Pristine 1T-$W_2N$ exhibits a symmetric octahedral coordination around the nitrogen atom with an equivalent W–N bond length of 2.1615 Å and an in-plane W–W distance of 3.3146 Å. The identical W–N bond lengths reflect the high structural symmetry of the pristine monolayer and indicate a uniform local bonding environment.

The substitution of one W atom by Ta forms the Janus 1T-WTaN monolayer and breaks the out-of-plane symmetry of the crystal. As a result, two different metal–nitrogen bond lengths are obtained, namely 2.1278 Å (W–N) and 2.1988 Å (Ta–N), while the W–Ta bond length is 3.2478 Å. The longer Ta–N bond compared with the W–N bond originates from the relatively larger atomic radius of tantalum, resulting in a slight distortion of the local coordination environment.

For the Janus 1T-WTaC monolayer, replacing nitrogen with carbon further modifies the local bonding environment. The optimized W–C and Ta–C bond lengths are calculated to be 2.0874 Å and 2.1347 Å, respectively, whereas the W–Ta distance decreases to 2.9080 Å. The shorter metal–carbon bonds indicate stronger covalent interaction between the transition-metal atoms and carbon compared with the corresponding metal–nitrogen bonds. These structural modifications influence the electronic structure, lattice dynamics, electron–phonon coupling, and consequently the superconducting properties of the investigated Janus monolayers.

Overall, Janus engineering introduces structural asymmetry by producing inequivalent

metal–anion bond lengths and modifying the metal–metal separation. These changes provide an effective route for tuning the physical and superconducting properties of two-dimensional tungsten nitride-based Janus monolayers.

**Table 4:** Optimized bond lengths of the investigated monolayers.

| System | Bond Length (Å) |
|---|---|
| 1T-$W_2N$ | W–N = 2.1615, W–W = 3.3146 |
| 1T-WTaN | W–N = 2.1278, Ta–N = 2.1988, W–Ta = 3.2478 |
| 1T-WTaC | W–C = 2.0874, Ta–C = 2.1347, W–Ta = 2.9080 |

## 5.4 Electronic Properties

A material's electrical conductivity, chemical bonding, and superconducting activity can all be inferred from its electronic characteristics. Electronic energy bands are created in crystalline solids by the interaction of atomic orbitals, and the density of states close to the Fermi level controls the strength of the electron-phonon interaction and the availability of charge carriers. Consequently, the electrical structure is essential to comprehending conventional superconductivity. This paper examines the impact of Janus engineering on the electronic properties of pristine 1T-$W_2N$, Janus 1T-WTaN, and Janus 1T-WTaC monolayers by analyzing their electronic band structures and projected density of states (PDOS).

### 5.4.1 Electronic Band Structure and Density of States of Pristine 1T-$W_2N$ Monolayer

The pristine 1T-$W_2N$ monolayer's electronic band structure and projected density of states (PDOS) are shown in Fig. 8. To examine the electrical characteristics of the optimized structure, the electronic band structure was computed along the high-symmetry path $\Gamma$–M–K–$\Gamma$.

The metallic character of pristine 1T-$W_2N$ is confirmed by multiple electronic bands crossing the Fermi level ($E_F = 0$ eV), as seen in Fig. 8. Good electrical conductivity is indicated by the lack of an energy band gap. The bands near the Fermi level exhibit both highly dispersive and relatively flat characteristics, suggesting the coexistence of charge carriers with different effective masses. Several nearly degenerate bands are also observed around the high-symmetry points due to the symmetry of the crystal structure.

The PDoS shows that tungsten atoms dominate the electronic states at the Fermi level, whereas the contribution from nitrogen atoms is minor. A significant DOS peak at

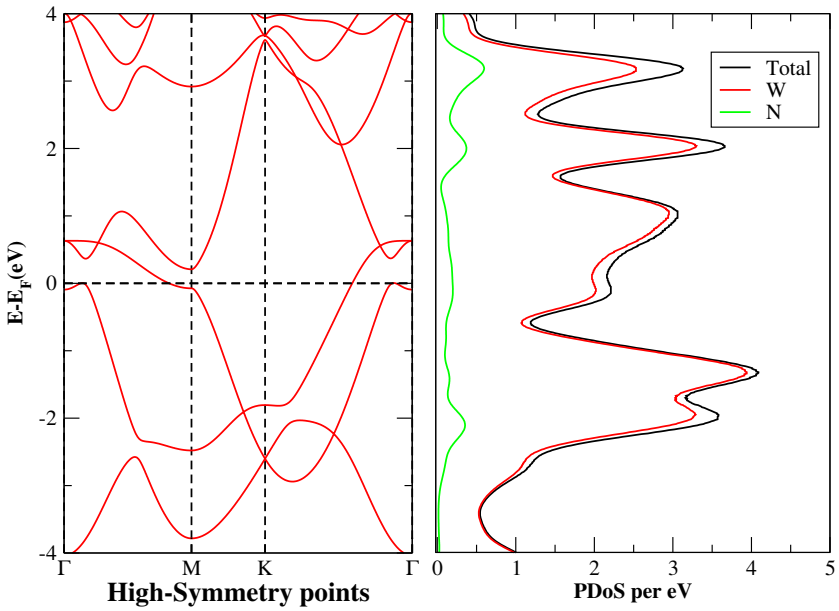



**Figure 8:** Calculated electronic band structure and projected density of states of the pristine 1T-$W_2N$ monolayer.

$-2.5$ eV is mainly composed of W atoms, while N atoms dominantly exist at the lower valence-band region. From the orbital-resolved PDOS, the W-$5d$ orbitals are primarily responsible for the electronic states at the Fermi level, while the contributions from W-$6s$ and W-$6p$ orbitals are negligible. The N-$2p$ orbitals dominantly participate in the formation of the valence-band states through hybridization with the W-$5d$ orbitals, implying the strong covalent character of W–N bonds.

The finite density of states of W-$5d$ at the Fermi level ensures the metallic nature of the pristine 1T-W $_2$N. The transition-metal $d$ states are identified as the primary source of the electron–phonon interaction, which is supported by the large electron–phonon coupling constant calculated in this work ($\lambda = 1.5101$). Consequently, this study highlights that the pristine 1T-$W_2N$ is a promising superconductor that can be stabilized by the conventional electron-phonon coupling mechanism.

### 5.4.2 Electronic Band Structure and Density of States of Janus 1T-WTaC Monolayer

Fig. 9 shows the electronic band structure and projected density of states (PDOS) of the Janus 1T-WTaC monolayer. To examine the electrical characteristics of the optimized Janus structure, the electronic band structure was computed along the high-symmetry

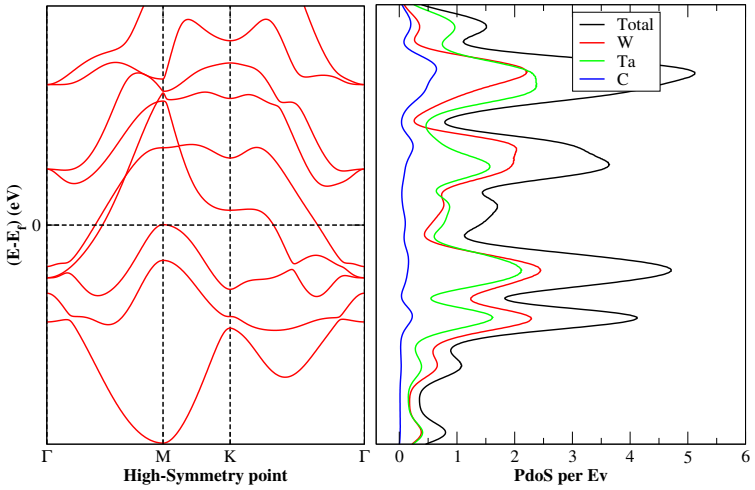



**Figure 9:** Calculated electronic band structure and projected density of states of the Janus 1T-WTaC monolayer.

path Γ–M–K–Γ.

A number of electronic bands pass the Fermi level ($E_{\mathrm{F}} = 0$ eV), as seen in Fig. 9, suggesting that the Janus 1T-WTaC monolayer is metallic. This monolayer appears to be a good conductor because there is no energy band gap. When one W is replaced by Ta and one N by C, the electronic band dispersion changes in comparison to the pristine 1T-$W_2N$. This change is caused by the broken out-of-plane symmetry, but the metallic character is still present.

The projected density of states indicates that transition-metal atoms make up the majority of the electronic states at the Fermi level, with carbon making up a small portion. The electronic structure around the Fermi level is primarily determined by the Ta-$5d$ orbitals, followed by the W-$5d$ orbitals, according to the orbital-resolved PDOS, while the contributions of the $s$ and $p$ orbitals are minimal. In Janus 1T-WTaC, significant covalent bonding results from the hybridization of C-$2p$ orbitals with W-$5d$ and Ta-$5d$ orbitals at the valence-band region.

The metallic nature of the Janus 1T-WTaC monolayer is guaranteed by the finite density of states of transition-metal $d$ at the Fermi level. The superconducting characteristics of the Janus monolayer, which will be covered below, are determined by the electronic states at the Fermi level, which directly engage in the electron–phonon interaction.

### 5.4.3 Electronic Band Structure and Density of States of Janus 1T-WTaN Monolayer

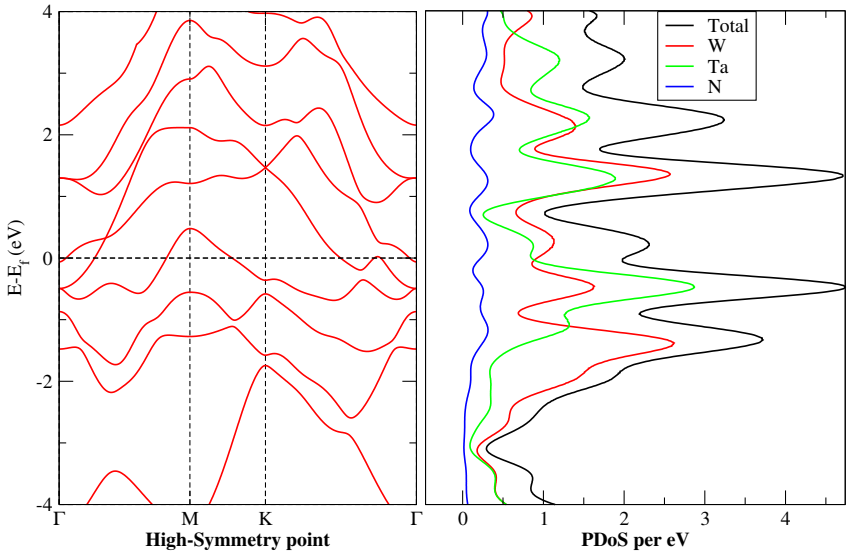



**Figure 10:** Calculated electronic band structure and projected density of states of the Janus 1T-WTaN monolayer.

The electronic band structure and projected density of states (PDOS) of the Janus 1T-WTaN monolayer are presented in Fig. 10. The electronic band structure was calculated along the high-symmetry path Γ–M–K–Γ to investigate the electronic properties of the optimized structure.

As shown in Fig. 10, several electronic bands cross the Fermi level ($E_{\mathrm{F}} = 0$ eV), confirming the metallic nature of the Janus 1T-WTaN monolayer. The absence of an energy band gap indicates that the material possesses good electrical conductivity. Compared with pristine 1T-$W_2N$, the replacement of one W layer by Ta breaks the out-of-plane mirror symmetry, resulting in slight modifications of the band dispersion while preserving the metallic electronic character.

The projected density of states further reveals that the electronic states near the Fermi level are mainly contributed by the transition-metal atoms (W and Ta), whereas the contribution from nitrogen is comparatively small. The orbital-resolved PDOS shows that the W-$5d$ and Ta-$5d$ orbitals dominate the states around the Fermi level, while the N-$2p$ orbitals mainly contribute to the valence-band region through hybridization with the transition-metal $d$ orbitals. This hybridization indicates strong W–N and Ta–N covalent bonding within the Janus structure.

The finite density of transition-metal $d$ states at the Fermi level confirms the metallic character of the Janus 1T-WTaN monolayer and provides the electronic states responsible for electron–phonon interaction. The modified electronic structure resulting from the Janus configuration plays an important role in determining the superconducting properties, which are discussed in the subsequent sections.

## 5.5 Dynamical Properties

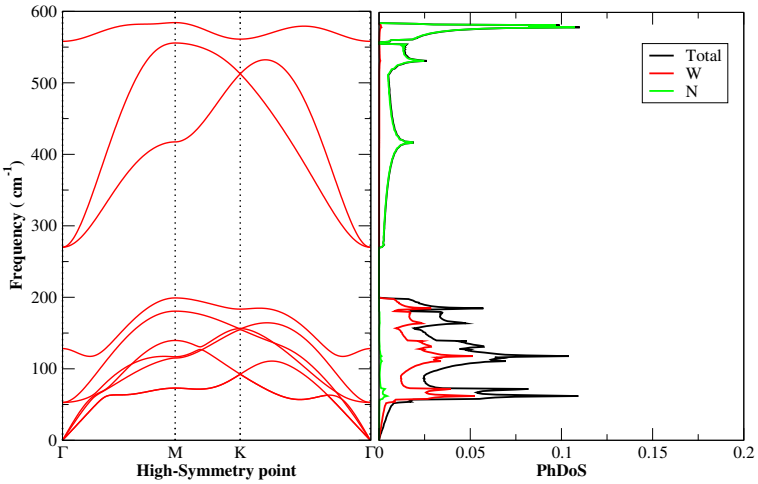
The dynamical properties of a material define the nature of the lattice vibrations. The phonon dispersion and projected phonon density of states (PhPDOS) determine the vibrational modes of the crystal and indicate the dynamical stability of the optimized structure by the absence of imaginary frequencies. The lattice vibrations are also responsible for the electron-phonon interaction, which is the key factor in conventional superconductivity. Thus, in this section, we present the phonon dispersion and PhPDOS of the pristine 1T-$W_2N$, Janus 1T-WTaN, and Janus 1T-WTaC monolayers to determine their dynamical stability and lattice vibrations. The calculated phonon spectra are further used to determine the electron-phonon coupling and superconducting properties in the following section.

### 5.5.1 Phonon Dispersion and Projected Phonon Density of States of Pristine 1T-$W_2N$

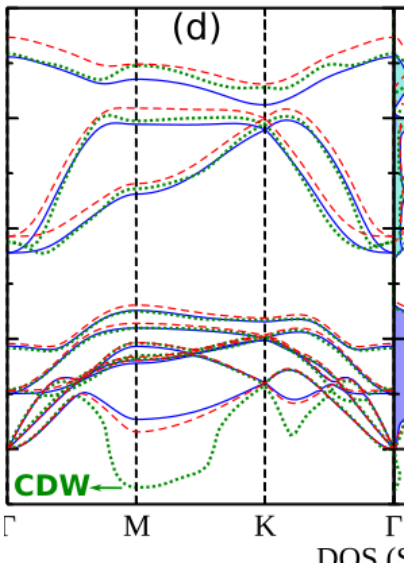
Fig. 11a shows the computed phonon dispersion and projected phonon density of states (PhPDOS) of the pristine 1T-$W_2N$ monolayer. Density functional perturbation theory (DFPT) was used to compute the phonon dispersion along the high-symmetry path $\Gamma$–M–K–$\Gamma$, and the estimated interatomic force constants were used to get the projected phonon density of states.

The optimized pristine 1T-$W_2N$ monolayer is dynamically stable since the phonon spectrum shows no imaginary (negative) frequencies across the whole Brillouin zone. The relaxed structure corresponds to a stable equilibrium state and is not prone to spontaneous lattice distortion, as indicated by the lack of soft phonon modes.

The phonon spectrum has nine vibrational branches, including three acoustic and six optical modes, because the primordial unit cell has three atoms. The crystal's translational invariance is satisfied by the longitudinal acoustic (LA), transverse acoustic (TA), and out-of-plane flexural acoustic (ZA) branches, which start at zero frequency at the $\Gamma$ point. A phonon gap between around 250 and 300 $cm^{-1}$ follows the acoustic branches,




**(a)** Present work




**(b)** Previous work

**Figure 11:** Comparison of the phonon dispersion and projected phonon density of states (PhPDOS) of the pristine 1T-$W_2N$ monolayer: (a) present work and (b) previous work.

which stretch up to about 180 $cm^{-1}$. Six optical phonon branches up to approximately 600 $cm^{-1}$ are seen above this gap.

Additional information on the atomic contributions to the lattice vibrations can be obtained from the projected phonon density of states. The heavier tungsten (W) atoms predominate in the low-frequency area below around 200 $cm^{-1}$. On the other hand, the lighter nitrogen (N) atoms are the primary source of the high-frequency optical modes between roughly 500 and 600 $cm^{-1}$. The intermediate-frequency band between roughly 300 and 450 $cm^{-1}$ is contributed to by both W and N atoms, suggesting linked W–N lattice vibrations resulting from the chemical bonding inside the monolayer.

The reported results are in good agreement with the overall phonon dispersion. The dynamical stability of pure 1T-$W_2N$ is confirmed by the current computation, which, in contrast to the prior analysis, shows no charge-density-wave (CDW) instability.

### 5.5.2 Phonon Dispersion and Projected Phonon Density of States of Janus 1T-WTaC

Fig. 12 displays the estimated phonon density of states (PhPDOS) and predicted phonon dispersion of the Janus 1T-WTaC monolayer. Density functional perturbation theory (DFPT) was used to determine the phonon dispersion along the high-symmetry path Γ–M–K–Γ, and the interatomic force constants were used to calculate the associated projected phonon density of states. The optimized Janus 1T-WTaC monolayer is dynamically stable since the phonon spectrum shows no imaginary (negative) frequencies across the whole Brillouin zone.. The absence of soft phonon modes indicates that the relaxed structure corresponds to a stable equilibrium configuration without any tendency toward spontaneous lattice distortion.

Since the primitive unit cell contains three atoms (W, Ta, and C), the phonon spectrum consists of nine vibrational branches, including three acoustic and six optical modes. The longitudinal acoustic (LA), transverse acoustic (TA), and flexural acoustic (ZA) branches originate from zero frequency at the Γ point, satisfying the translational invariance of the crystal. The acoustic branches extend up to approximately 180 $cm^{-1}$, followed by a wide phonon gap extending from nearly 200 to 650 $cm^{-1}$. Above this gap, the optical phonon branches are concentrated within the frequency range of approximately 660–700 $cm^{-1}$, with the highest optical mode reaching nearly 700 $cm^{-1}$. The projected phonon density of states provides further insight into the atomic contributions to the vibrational modes. The low-frequency acoustic region below approximately 120 $cm^{-1}$ is dominated primarily by the heavier tantalum (Ta) atoms, while tungsten (W) also contributes to this region with comparatively lower intensity. Between approx-

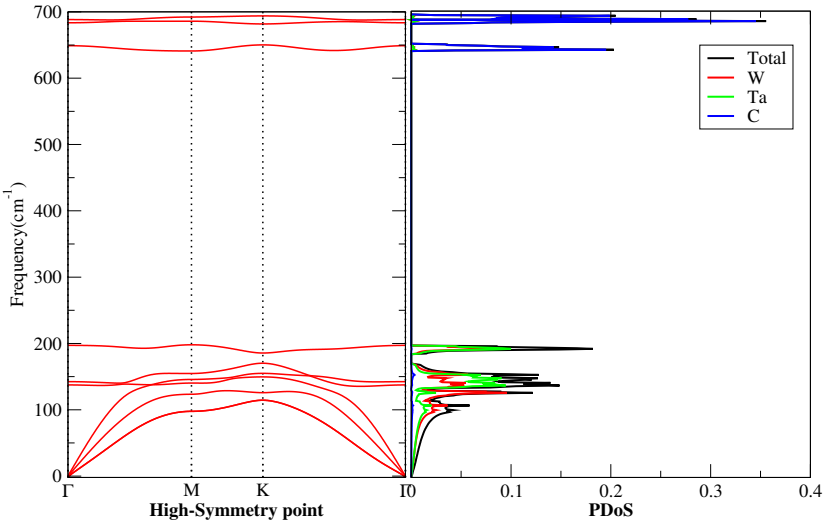



**Figure 12:** Calculated phonon dispersion and projected phonon density of states (PhPDOS) of the pristine 1T-$W_2N$ monolayer.

imately 120 and 200 cm$^{-1}$, both W and Ta atoms contribute significantly, indicating coupled vibrations of the transition-metal layers. A wide phonon gap extending from about 200 to 650 cm$^{-1}$ is observed, where virtually no phonon states are present. Above this gap, the optical modes appear in the frequency range of approximately 650–710 cm$^{-1}$ and are dominated almost entirely by the carbon (C) atoms. The highest optical peak, located near 700 cm$^{-1}$, originates mainly from C-related bond-stretching vibrations, whereas the contributions from W and Ta are negligible in this region. The clear separation between the transition-metal and carbon vibrations reflects the significant mass difference between these atoms and gives rise to the pronounced phonon gap observed in the dispersion spectrum.

### 5.5.3 Phonon Dispersion and Projected Phonon Density of States of Janus 1T-WTaN Monolayer

Fig. 13 displays the estimated phonon density of states (PhPDOS) and computed phonon dispersion of the Janus 1T-WTaN monolayer. Density functional perturbation theory (DFPT) was used to derive the phonon dispersion along the high-symmetry path Γ–M–K–Γ, and the accompanying interatomic force constants were used to compute the projected phonon density of states.

The phonon dispersion of the optimized Janus 1T-WTaN monolayer exhibits a small imaginary phonon frequency of approximately $-6$ cm$^{-1}$ near the Γ point, indicating

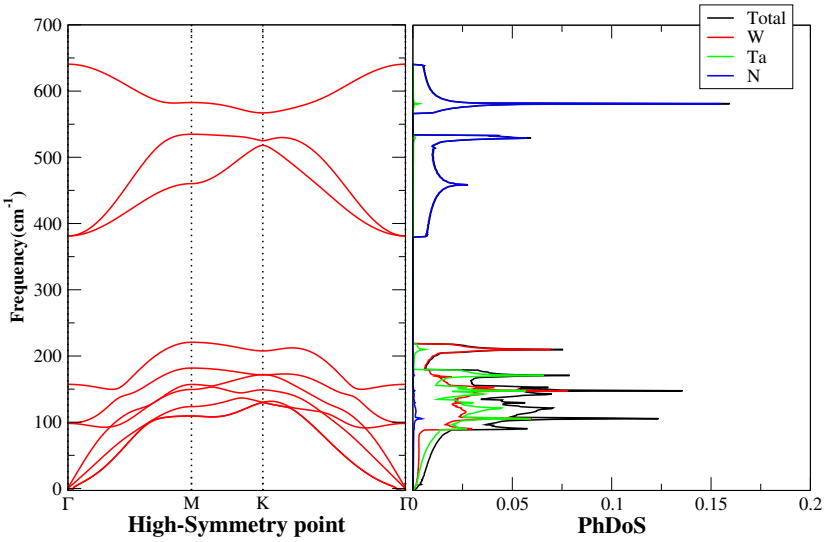



**Figure 13:** Calculated phonon dispersion and projected phonon density of states (PhPDOS) of the Janus 1T-WTaN monolayer.

a weak dynamical instability. The magnitude of this negative frequency is very small and is generally associated with soft out-of-plane vibrational modes that are frequently observed in two-dimensional materials. Such a small imaginary mode suggests that the structure is very close to a dynamically stable configuration and can often be stabilized through external perturbations such as mechanical strain. Therefore, tensile strain was introduced in the present work to investigate whether the lattice instability could be suppressed. After applying a 1% tensile strain, the imaginary phonon frequency decreased to approximately $-2.85$ cm$^{-1}$, demonstrating that tensile strain effectively improves the dynamical stability of the Janus monolayer. A detailed discussion of the strain-dependent phonon spectrum and superconducting properties is presented in the subsequent sections.

Since the primitive unit cell contains three atoms (W, Ta, and N), the phonon spectrum consists of nine vibrational branches, including three acoustic and six optical modes. The longitudinal acoustic (LA), transverse acoustic (TA), and flexural acoustic (ZA) branches originate from zero frequency at the $\Gamma$ point, satisfying the translational invariance of the crystal. The acoustic branches extend to approximately 220 cm$^{-1}$, followed by a phonon gap between about 220 and 420 cm$^{-1}$. Above this gap, the six optical phonon branches are distributed between approximately 420 and 650 cm$^{-1}$, with the highest optical mode reaching nearly 650 cm$^{-1}$.

The projected phonon density of states provides detailed information regarding the atomic origin of the vibrational modes. The low-frequency region below approximately

100 $cm^{-1}$ is mainly dominated by the heavier tungsten (W) and tantalum (Ta) atoms, whereas the nitrogen (N) atom contributes very little in this region. Between approximately 100 and 220 $cm^{-1}$, all three atomic species participate in the lattice vibrations, indicating coupled W–Ta–N vibrational modes, with the Ta contribution being slightly stronger than that of W.Between between 220 and 420 $cm^{-1}$, a clear phonon gap is seen, where virtually no vibrational modes exist. The nitrogen atoms dominate the phonon density of states in the high-frequency optical range between roughly 420 and 650 $cm^{-1}$, with contributions from W and Ta becoming insignificant. The stretching vibrations of the W–N and Ta–N bonds are primarily responsible for the strongest peak, which appears at 500 $cm^{-1}$. The significant mass differential between the heavy W/Ta atoms and the significantly lighter nitrogen atom is the source of the distinct distinction between the transition-metal dominated acoustic modes and the nitrogen-dominated optical modes.

Overall, the phonon dispersion and projected phonon density of states reveal that the unstrained Janus 1T-WTaN monolayer is only weakly dynamically unstable because of the small imaginary phonon mode near the $\Gamma$ point. The reduction of this imaginary frequency under tensile strain indicates that strain engineering is an effective approach for stabilizing the Janus structure, thereby providing a suitable platform for investigating its electron–phonon interaction and superconducting properties.

## 5.6 Eliashberg Spectral Function ($\alpha^2F(\omega)$)

The superconducting properties of conventional superconductors are determined by the interaction of conduction electrons with lattice vibrations, phonons. According to Migdal–Eliashberg theory, this interaction can be described in terms of the Eliashberg spectral function, $\alpha^2F(\omega)$, which determines the electron–phonon coupling and provides a microscopic understanding of the mechanism of superconductivity. In contrast to the Bardeen–Cooper–Schrieffer (BCS) theory of superconductivity, which uses a simplified description of the interaction, the Eliashberg theory takes into account the coupling's dependence on the phonon frequency, providing a more accurate description of superconducting materials' properties.

Therefore, it is possible to identify the phonon frequencies that cause the electron-phonon interaction by combining the information about the electron-phonon matrix elements and the phonon density of states in the Eliashberg function. Therefore, the coupling constant ($\lambda$) and the logarithmic average frequency ($\omega_{\log}$)) may be computed using the Eliashberg spectral function. The superconducting transition temperature ($T_c$)) can then be estimated using the Allen–Dynes modified McMillan formula. In this study, the

Eliashberg spectral function was calculated for pristine 1T-$W_2N$ as well as for the Janus 1T-WTaN and 1T-WTaC monolayers. Using the calculated $\alpha^2F(\omega)$, electron–phonon coupling constant, and phonon spectra, we analyze how substitution of transition-metal elements and structural asymmetry in Janus materials influence the electron–phonon interaction and superconducting properties.

### 5.6.1 Eliashberg Spectral Function and Electron–Phonon Coupling Constant of Pristine 1T-$W_2N$ Monolayer

The calculated Eliashberg spectral function, $\alpha^2F(\omega)$, and the cumulative electron–phonon coupling constant, $\lambda(\omega)$, for the pristine 1T-$W_2N$ monolayer are presented in Fig. 14. The Eliashberg spectral function describes the contribution of phonons with different frequencies to the electron–phonon interaction, whereas the cumulative electron–phonon coupling constant illustrates how the total coupling strength builds up with increasing phonon frequency.

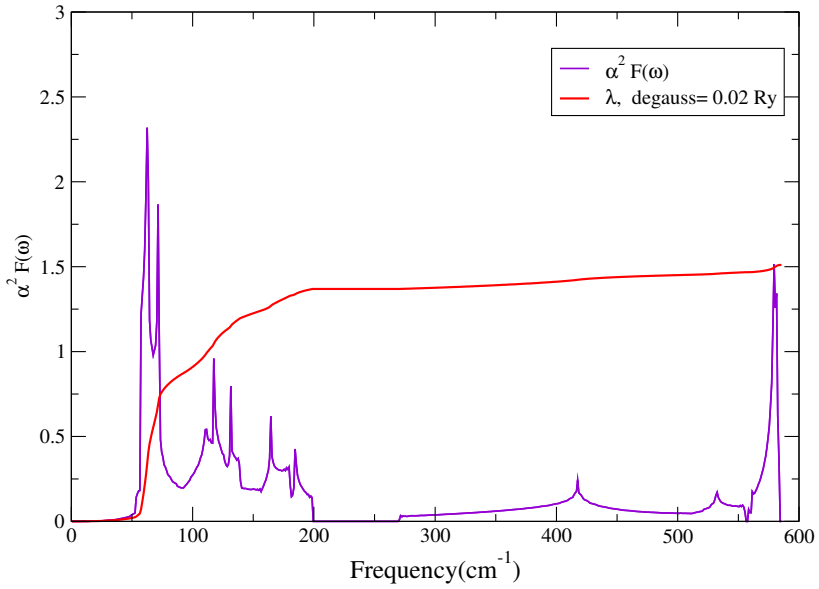



**Figure 14:** Calculated Eliashberg spectral function $\alpha^2F(\omega)$ and cumulative electron–phonon coupling constant $\lambda(\omega)$ of the pristine 1T-$W_2N$ monolayer.

As shown in Fig. 14, the Eliashberg spectral function exhibits several pronounced peaks over the entire phonon frequency range, indicating that different phonon modes contribute to the electron–phonon interaction. The most intense peaks are observed in the low-frequency region between approximately 60 and 90 $cm^{-1}$, revealing that low-energy phonons provide the dominant contribution to the electron–phonon coupling. Additional

smaller peaks appear within the intermediate-frequency range below about 200 $cm^{-1}$, suggesting that these phonon modes also participate in the coupling process. In contrast, only weak peaks are present in the high-frequency region around 430 $cm^{-1}$ and 530–560 $cm^{-1}$.

The cumulative electron–phonon coupling constant, $\lambda(\omega)$, increases rapidly within the low-frequency region and accumulates most of its final value below approximately 200 $cm^{-1}$. Beyond this frequency, the growth of $\lambda(\omega)$ becomes much slower and gradually approaches saturation above approximately 250 $cm^{-1}$. This behavior indicates that the low-frequency phonons are primarily responsible for the superconducting pairing interaction, whereas the contribution from high-frequency optical phonons is comparatively small.

Although several optical phonon modes are observed at higher frequencies, they produce only a slight increase in the cumulative electron–phonon coupling constant, indicating relatively weak coupling with the electronic states near the Fermi level. These high-frequency modes mainly originate from the vibrations of the lighter nitrogen atoms, while the dominant contribution to the electron–phonon interaction arises from the low-frequency vibrations associated with the heavier tungsten atoms.

The calculated total electron–phonon coupling constant for pristine 1T-$W_2N$ is found to be

$$\lambda = 1.5101,$$

which indicates a relatively strong electron–phonon interaction. Since the majority of the coupling originates from low-frequency phonons, these lattice vibrations play the dominant role in the formation of Cooper pairs. The comparatively large value of $\lambda$ suggests that pristine 1T-$W_2N$ is a promising conventional phonon-mediated superconductor and provides a reliable foundation for evaluating its superconducting transition temperature using the Allen–Dynes modified McMillan equation.

### 5.6.2 Eliashberg Spectral Function and Electron–Phonon Coupling Constant of 1% Strained Janus 1T-WTaN Monolayer

Fig. 15 displays the computed Eliashberg spectral function, $\alpha^2F(\omega)$, and the cumulative electron–phonon coupling constant, $\lambda(\omega)$, for the 1% biaxially strained Janus 1T-WTaN monolayer. The cumulative coupling constant shows how phonons at various frequencies contribute to the overall electron-phonon coupling, whereas the Eliashberg spectral function gives the phonon-frequency-dependent strength of the electron-phonon interaction.

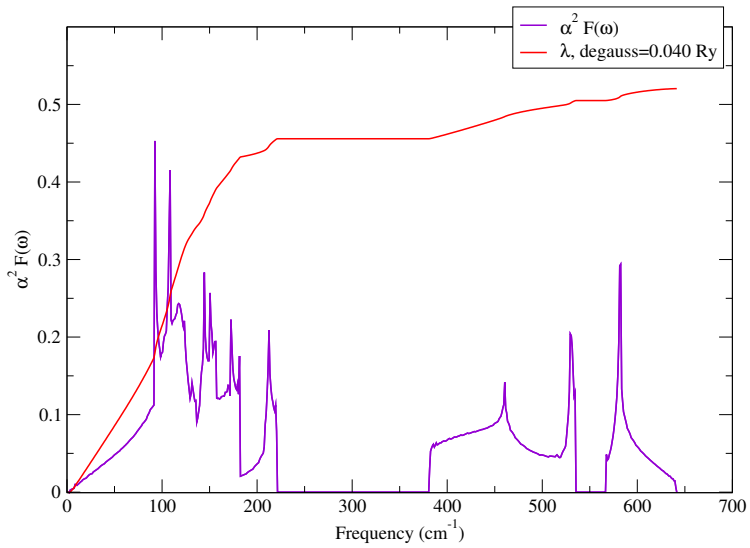



**Figure 15:** Calculated Eliashberg spectral function $\alpha^2F(\omega)$ and cumulative electron–phonon coupling constant $\lambda(\omega)$ of the 1% strained Janus 1T-WTaN monolayer.

As shown in Fig. 15, the Eliashberg spectral function exhibits several distinct peaks distributed over the phonon frequency range, indicating that different phonon modes contribute to the electron–phonon interaction. The largest spectral weight is concentrated in the low-frequency region between approximately 80 and 180 $\text{cm}^{-1}$, where the strongest peaks of $\alpha^2F(\omega)$ are observed. These low-energy phonons provide the dominant contribution to the overall electron–phonon coupling. Additional smaller peaks are present between approximately 180 and 250 $\text{cm}^{-1}$, whereas only weak contributions are observed in the high-frequency optical region around 430–620 $\text{cm}^{-1}$.

The cumulative electron–phonon coupling constant increases rapidly within the low-frequency region and reaches most of its final value below approximately 220 $\text{cm}^{-1}$. Beyond this frequency, the increase in $\lambda(\omega)$ becomes much slower and gradually approaches saturation above about 300 $\text{cm}^{-1}$. This behavior demonstrates that the superconducting pairing interaction is governed mainly by low-frequency lattice vibrations, while the high-frequency optical phonons contribute only marginally to the total coupling strength.

The high-frequency peaks observed between approximately 430 and 620 $\text{cm}^{-1}$ mainly originate from the nitrogen-dominated optical vibrations identified in the phonon density of states. Although these optical modes possess relatively high vibrational frequencies, they produce only a small increase in the cumulative coupling constant, indicating relatively weak electron–phonon matrix elements. Consequently, the low-frequency vibrations associated primarily with the heavier W and Ta atoms play the dominant role

in the electron–phonon interaction responsible for superconductivity.

The calculated cumulative electron–phonon coupling constant converges to

$$\lambda = 0.5204,$$

which is substantially smaller than that of pristine 1T-$W_2N$ ($\lambda = 1.5101$). The reduction in the coupling strength can be attributed to the combined effects of Janus symmetry breaking and W→Ta substitution, which modify both the lattice dynamics and the electron–phonon interaction. Although the 1% strained Janus 1T-WTaN monolayer remains metallic and dynamically stable, the weaker electron–phonon coupling suggests a lower superconducting transition temperature than that of pristine 1T-$W_2N$.

### 5.6.3 Eliashberg Spectral Function and Electron–Phonon Coupling Constant of Janus 1T-WTaC Monolayer

The calculated Eliashberg spectral function, $\alpha^2F(\omega)$, together with the cumulative electron–phonon coupling constant, $\lambda(\omega)$, for the Janus 1T-WTaC monolayer is presented in Fig. 16. The Eliashberg spectral function describes the contribution of phonons with different frequencies to the electron–phonon interaction, whereas the cumulative coupling constant illustrates the evolution of the total electron–phonon coupling with increasing phonon frequency.

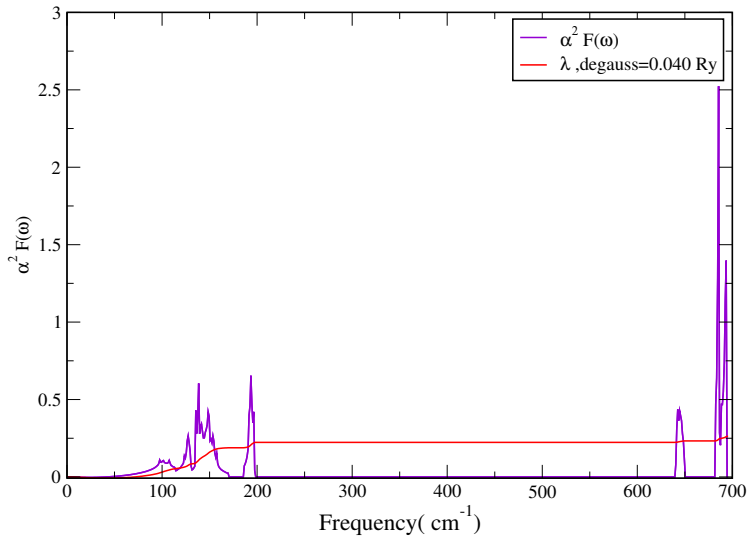



**Figure 16:** Calculated Eliashberg spectral function $\alpha^2F(\omega)$ and cumulative electron–phonon coupling constant $\lambda(\omega)$ of the Janus 1T-WTaC monolayer.

As shown in Fig. 16, the Eliashberg spectral function exhibits several peaks distributed over the phonon spectrum, indicating that different vibrational modes contribute to the electron–phonon interaction. The largest contribution is concentrated in the low-frequency region between approximately 80 and 180 $cm^{-1}$, where the strongest peaks of $\alpha^2F(\omega)$ are observed. Smaller contributions are also present in the intermediate-frequency range below about 250 $cm^{-1}$. In contrast, the high-frequency optical region extending from approximately 600 to 690 $cm^{-1}$ contains relatively weak contributions to the electron–phonon interaction.

The cumulative electron–phonon coupling constant increases rapidly within the low-frequency region and reaches most of its final value below approximately 220 $cm^{-1}$. Above this frequency, the increase in $\lambda(\omega)$ becomes very gradual and approaches saturation near 300 $cm^{-1}$. This behavior indicates that the low-frequency phonons make the dominant contribution to the total electron–phonon coupling, whereas the high-frequency optical phonons contribute only slightly despite their relatively large vibrational energies.

The high-frequency peaks observed between approximately 600 and 690 $cm^{-1}$ are mainly associated with the carbon-dominated optical vibrations. Although these optical modes possess considerably higher frequencies than the acoustic modes, they produce only a small increase in the cumulative electron–phonon coupling constant, indicating relatively weak coupling with the electronic states near the Fermi level. Consequently, the superconducting pairing interaction in the Janus 1T-WTaC monolayer is governed primarily by the low-frequency vibrations of the heavier W and Ta atoms.

The calculated cumulative electron–phonon coupling constant converges to

$$\lambda = 0.2622,$$

which is substantially smaller than those obtained for pristine 1T-$W_2N$ ($\lambda = 1.5101$) and Janus 1% strained 1T-WTaN ($\lambda = 0.5204$). The significant reduction in the electron–phonon coupling strength indicates that replacing nitrogen with carbon weakens the phonon-mediated pairing interaction. As a result, although the Janus 1T-WTaC monolayer remains metallic, its comparatively small electron–phonon coupling suggests a much lower superconducting transition temperature than those of pristine 1T-$W_2N$ and Janus 1T-WTaN.

## 5.7 Superconducting Transition Temperature ($T_c$)

The temperature below which a material transitions from the normal conducting state to the superconducting state via the creation of Cooper pairs mediated by electron–phonon interaction is known as the superconducting transition temperature ($T_c$). The electron–phonon coupling constant ($\lambda$), the logarithmic average phonon frequency ($\omega_{\log}$), and the Coulomb pseudopotential ($\mu^*$) are the primary factors influencing the superconducting transition temperature in the Migdal–Eliashberg formalism. The Allen–Dynes modified McMillan equation is used in this study to determine the superconducting transition temperatures of pristine 1T-$W_2N$, 1% strained Janus 1T-WTaN, and Janus 1T-WTaC monolayers. To comprehend the impact of biaxial tensile strain and Janus engineering on the superconducting characteristics of these two-dimensional monolayers, the computed $T_c$ values are examined.

### 5.7.1 Superconducting Transition Temperature ($T_c$) of Pristine 1T-$W_2N$ Monolayer

The superconducting transition temperature ($T_c$) of the pristine 1T-$W_2N$ monolayer was calculated using the Allen–Dynes modified McMillan equation based on the computed electron–phonon coupling constant ($\lambda$), logarithmic average phonon frequency ($\omega_{\log}$), and the effective Coulomb pseudopotential ($\mu^*$). The variation of $T_c$ with $\mu^*$ for a Gaussian broadening of 0.05 Ry is shown in Fig. 17. The superconducting parameters obtained from the electron–phonon calculations are summarized in Table 5.

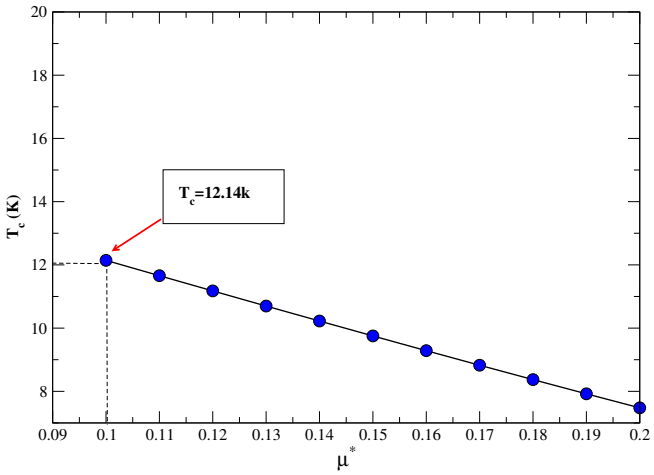



**Figure 17:** Variation of the superconducting transition temperature ($T_c$) with Coulomb pseudopotential ($\mu^*$) for the pristine 1T-$W_2N$ monolayer.

Figure 17 shows that the superconducting transition temperature decreases continuously with increasing Coulomb pseudopotential. As the value of $\mu^*$ increases, the screened electron–electron Coulomb repulsion becomes stronger, which weakens the phonon-mediated Cooper-pair interaction and consequently reduces the superconducting transition temperature. For the commonly adopted value of $\mu^* = 0.10$, the calculated transition temperature is 12.14 K. As $\mu^*$ increases to 0.20, the transition temperature decreases gradually to approximately 7.48 K. Although the Allen–Dynes equation predicts an exponential dependence of $T_c$ on $\mu^*$, the variation appears nearly linear within the narrow range of $\mu^*$ (0.10–0.20) considered in the present work.

The relatively large electron–phonon coupling constant ($\lambda = 1.5101$) obtained for pristine 1T-$W_2N$ leads to a comparatively high superconducting transition temperature among the investigated systems. These results indicate that pristine 1T-$W_2N$ is a promising conventional two-dimensional superconductor and provide a reliable reference for evaluating the effects of Janus engineering and strain on the superconducting properties of the WTaN and WTaC monolayers.

**Table 5:** Calculated electron–phonon coupling constant ($\lambda$), logarithmic average phonon frequency ($\omega_{\log}$), superconducting transition temperature ($T_c$), and Gaussian broadening parameter for pristine $W_2N$ monolayer ($\mu^* = 0.10$).

| **Broadening (Ry)** | $\boldsymbol{\lambda}$ | $\boldsymbol{\omega_{log}}$ **(K)** | $\mathbf{T_c}$ **(K)** |
|---|---|---|---|
| 0.005 | 1.6999 | 127.9670 | 17.296 |
| 0.010 | 1.6331 | 136.9496 | 17.053 |
| 0.015 | 1.5487 | 140.4078 | 16.592 |
| 0.020 | 1.5004 | 141.5284 | 16.138 |
| 0.025 | 1.4403 | 141.9219 | 15.414 |
| 0.030 | 1.3718 | 142.1712 | 14.534 |
| 0.035 | 1.3051 | 142.3968 | 13.668 |
| 0.040 | 1.2453 | 142.6177 | 12.881 |
| 0.045 | 1.1937 | 142.8500 | 12.237 |
| 0.050 | 1.1500 | 143.1130 | 12.142 |

### 5.7.2 Superconducting Transition Temperature ($T_c$) of 1% Strained Janus 1T-WTaN Monolayer

The Allen–Dynes modified McMillan equation was used to determine the superconducting transition temperature ($T_c$) of the 1% strained Janus 1T-WTaN monolayer based on the computed electron–phonon coupling constant ($\lambda$), logarithmic average phonon frequency ($\omega_{\log}$), and the effective Coulomb pseudopotential ($\mu^*$). Fig. 18 illustrates how $T_c$

changes with $\mu^*$ for a Gaussian broadening of 0.05 Ry. The calculated superconducting parameters obtained from the electron–phonon calculations are summarized in Table 6.

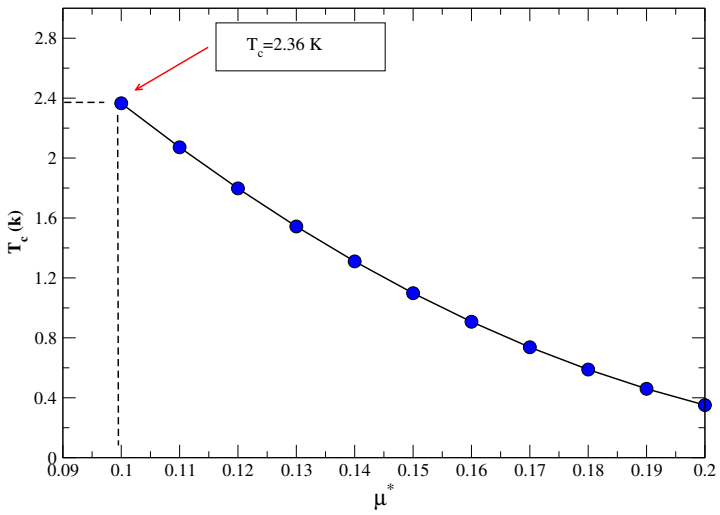



**Figure 18:** Variation of the superconducting transition temperature ($T_c$) with Coulomb pseudopotential ($\mu^*$) for the 1% strained Janus 1T-WTaN monolayer.

Figure 18 illustrates the variation of the superconducting transition temperature with the Coulomb pseudopotential. As the value of $\mu^*$ increases, the screened electron–electron Coulomb repulsion becomes stronger, suppressing the phonon-mediated Cooper-pair interaction and leading to a continuous decrease in the superconducting transition temperature. For the commonly adopted value of $\mu^* = 0.10$, the calculated transition temperature is 2.37 K. As $\mu^*$ increases to 0.20, the transition temperature decreases gradually to approximately 0.35 K. Although the Allen–Dynes equation predicts an exponential dependence of $T_c$ on $\mu^*$, the variation appears nearly linear over the relatively narrow range of $\mu^*$ (0.10–0.20) considered in this work.

Compared with pristine 1T-$W_2N$, the 1% strained Janus 1T-WTaN monolayer exhibits a significantly lower superconducting transition temperature because of its reduced electron–phonon coupling strength ($\lambda = 0.5204$). The combined effects of Janus symmetry breaking and tensile strain modify the lattice dynamics and weaken the electron–phonon interaction, resulting in a lower $T_c$. Nevertheless, the calculated transition temperature confirms that the strained Janus 1T-WTaN monolayer retains conventional phonon-mediated superconductivity.

**Table 6:** Calculated EPC constant ($\lambda$), logarithmic average phonon frequency ($\omega_{\log}$), superconducting transition temperature ($T_c$), and Gaussian broadening parameter for Janus WTaN monolayer ($\mu^* = 0.10$).

| Broadening (Ry) | $\lambda$ | $\omega_{\log}$ (K) | $T_c$ (K) |
|---|---|---|---|
| 0.005 | 0.8084 | 227.9610 | 12.039 |
| 0.010 | 0.6326 | 196.1413 | 4.229 |
| 0.015 | 0.6034 | 168.9319 | 3.251 |
| 0.020 | 0.5732 | 154.9384 | 2.579 |
| 0.025 | 0.5469 | 147.7694 | 2.156 |
| 0.030 | 0.5312 | 144.3023 | 1.969 |
| 0.035 | 0.5248 | 142.9652 | 1.904 |
| 0.040 | 0.5256 | 142.8395 | 1.914 |
| 0.045 | 0.5319 | 143.3044 | 2.050 |
| 0.050 | 0.5421 | 143.9630 | 2.366 |

### 5.7.3 Superconducting Transition Temperature ($T_c$) of Janus 1T-WTaC Monolayer

The superconducting transition temperature ($T_c$) of the Janus 1T-WTaC monolayer was calculated using the Allen–Dynes modified McMillan equation based on the computed electron–phonon coupling constant ($\lambda$), logarithmic average phonon frequency ($\omega_{\log}$), and the effective Coulomb pseudopotential ($\mu^*$). The variation of $T_c$ with $\mu^*$ for a Gaussian broadening of 0.05 Ry is shown in Fig. 19. The calculated superconducting parameters are summarized in Table 7.

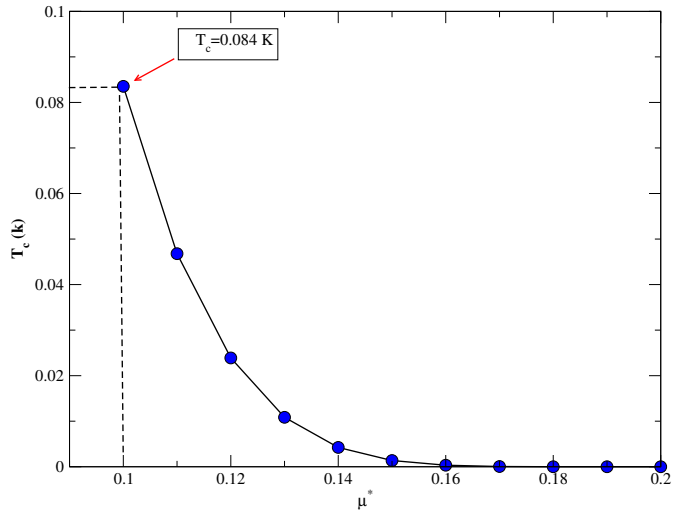



**Figure 19:** Variation of the superconducting transition temperature ($T_c$) with Coulomb pseudopotential ($\mu^*$) for the Janus 1T-WTaC monolayer.

Figure 19 shows that the superconducting transition temperature decreases rapidly with increasing Coulomb pseudopotential. As the value of $\mu^*$ increases, the screened electron–electron Coulomb repulsion suppresses the phonon-mediated Cooper-pair interaction, leading to a significant reduction in $T_c$. For the commonly adopted value of $\mu^* = 0.10$, the calculated transition temperature is only 0.0835 K. The transition temperature decreases sharply with increasing $\mu^*$ and becomes nearly zero for $\mu^*$ values of approximately 0.19–0.20, indicating that superconductivity is almost completely suppressed within this range.

The extremely low superconducting transition temperature is a direct consequence of the weak electron–phonon coupling strength ($\lambda = 0.2622$) obtained for the Janus 1T-WTaC monolayer. Compared with pristine 1T-$W_2N$ and the 1% strained Janus 1T-WTaN monolayer, the replacement of nitrogen by carbon substantially weakens the electron–phonon interaction, resulting in a much lower superconducting transition temperature. These results indicate that, although the Janus 1T-WTaC monolayer remains metallic, its superconducting behavior is significantly weaker than that of the other investigated systems.

**Table 7:** Calculated electron–phonon coupling constant ($\lambda$), logarithmic average phonon frequency ($\omega_{\log}$), superconducting transition temperature ($T_c$), and Gaussian broadening parameter for Janus WTaC monolayer ($\mu^* = 0.10$).

| **Broadening (Ry)** | $\boldsymbol{\lambda}$ | $\boldsymbol{\omega_{\log}}$ **(K)** | $\mathbf{T_c}$ **(K)** |
|---|---|---|---|
| 0.005 | 0.3384 | 262.5306 | 0.147 |
| 0.010 | 0.2279 | 291.6499 | 0.018 |
| 0.015 | 0.2187 | 283.3618 | 0.013 |
| 0.020 | 0.2238 | 273.1422 | 0.015 |
| 0.025 | 0.2326 | 268.2155 | 0.021 |
| 0.030 | 0.2428 | 265.7408 | 0.030 |
| 0.035 | 0.2538 | 263.4891 | 0.042 |
| 0.040 | 0.2654 | 260.5432 | 0.058 |
| 0.045 | 0.2773 | 256.8194 | 0.078 |
| 0.050 | 0.2891 | 252.6097 | 0.084 |

# Chapter 6

# Conclusion and recommendation

## 6.1 Conclusion

In this thesis, the structural, electronic, dynamical and superconducting properties of the 1T-W $_2$ N monolayer, Janus 1T-WTaN and Janus 1T-WTaC monolayers are systematically studied, using ab initio calculations based on density functional theory (DFT) and Density Functional Perturbation Theory (DFPT) as implemented in the Quantum ESPRESSO package. Norm-conserving PAW pseudopotentials and PBE exchange-correlation functional within generalized gradient approximation (GGA) are used. After convergence tests and geometry optimizations, we found that the lattice constant of the pristine 1T-W $_2$ N monolayer, Janus 1T-WTaN and Janus 1T-WTaC monolayers are about 2.80 A, 2.90 A and 3.10 A, respectively.

Our calculation results show that all the three materials have metallic character due to the presence of electron bands crossing the Fermi level. The orbital projected density of states indicates that the transition-metal $d$ orbitals dominate the electronic states near the Fermi energy while the N-$2p$ and C-$2p$ orbitals are mostly located at the valence-band maximum, with contributions peaked around the Fermi level due to hybridization.

The phonon dispersion calculations confirm the dynamic stability of the pristine 1T-W $_2$ N and Janus 1T-WTaC. In contrast, for the optimized Janus 1T-WTaN monolayer, a small imaginary frequency was found, which disappears under 1% biaxial tensile strain, thereby stabilizing the structure.

The calculated electron–phonon coupling constants are $\lambda = 1.5101$, $\lambda = 0.5204$, and $\lambda = 0.2622$ for pristine 1T-W $_2$ N, 1% strained Janus 1T-WTaN, and Janus 1T-WTaC, respectively. Using the Allen–Dynes modified McMillan equation with $\mu = 0.10$, the superconducting transition temperatures were estimated to be 12.14, 2.37, and 0.0835

K, respectively. Thus, the superconducting transition temperature decreases with decreasing electron–phonon coupling strength.

Overall, the study shows that the clean 1T-$W_2N$ is the best superconductor of all considered materials due to the strong electron–phonon coupling and a relatively high superconducting transition temperature. The Janus design reduces the electron–phonon coupling strength and, therefore, the superconducting transition temperature, although the metallic character of the monolayers is preserved. However, the stabilization of Janus 1T-WTaN by biaxial tensile strain opens the possibility to tune the structural stability and superconducting properties of Janus two-dimensional materials by strain engineering.

## 6.2 Future Outlook

Using first-principles calculations, the current work offers a theoretical explanation of the structural, electrical, and superconducting properties of pristine 1T-$W_2N$, Janus 1T-WTaN, and Janus 1T-WTaC monolayers. An $18\times18\times1$ k-point mesh, a $3\times3\times1$ q-point grid, and a plane-wave energy cutoff of 90 Ry were used for the computations in this work. Denser k-point and q-point meshes could be used in future studies to increase the precision of the computed electronic, phonon, and superconducting characteristics. The present study can be extended in several directions. First of all, it is worth using the Electron - Phonon Wannier (EPW) code to calculate a more accurate electron - phonon coupling and superconducting properties via Wannier interpolation. The effect of a larger biaxial or uniaxial strain, as well as the action of a hydrostatic pressure and an external electric field, can also be considered in order to tune T c . Moreover, surface adsorption and chemical functionalization, as well as the search for other Janus transition-metal nitrides and carbides, may be performed to enhance superconductivity. Last but not least, the present theoretical results may stimulate experimental synthesis and characterization of Janus WTaN and WTaC monolayers, which will open the way for exploring two-dimensional superconducting systems for future applications in nanoelectronics.

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